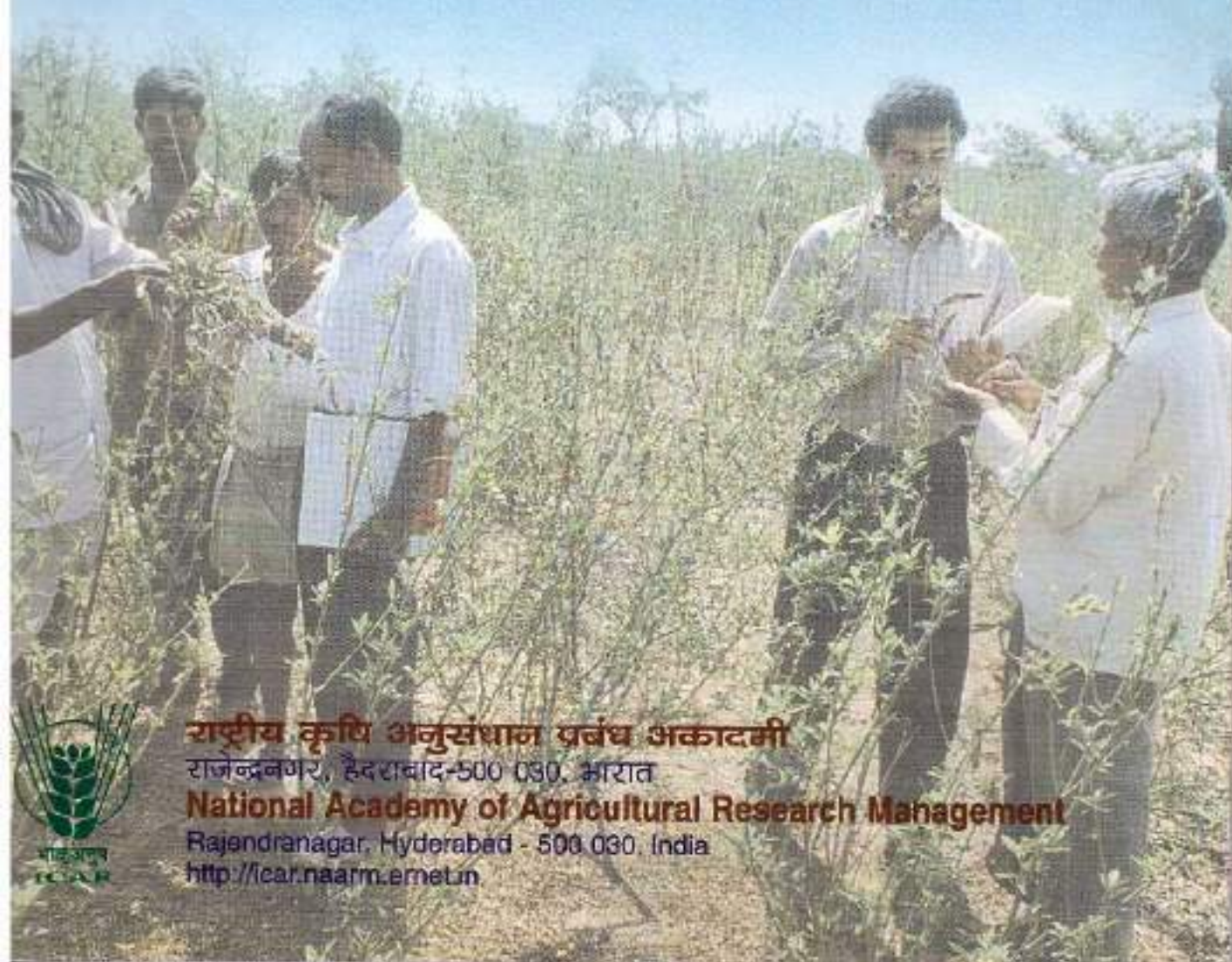


वार्षिक प्रतिवेदन Annual Report 2000-2001



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

राजेन्द्रनगर, हैदराबाद-500 030, भारत

National Academy of Agricultural Research Management

Rajendranagar, Hyderabad - 500 030, India

<http://icar.naarm.emet.in>



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Preface

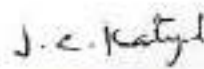
It gives me immense pleasure in presenting this document as a comprehensive record of the Academy's activities and accomplishments during 2000-2001. Since its inception, NAARM has taken up the onerous task of developing the vast human resource engaged in agricultural research, education and technology transfer functions in the country. As a participant in the process of bringing about change in attitude, value system, and commitment in agricultural scientists, NAARM has played a significant role by conducting training programmes, seminars, workshops, and interactive sessions. The year witnessed a spurt in the international training programmes. The international collaboration with Sri Lanka, Tanzania, Yemen and, ISNAR, the Netherlands provided excellent experience, which led to a positive impact on the Academy's performance.

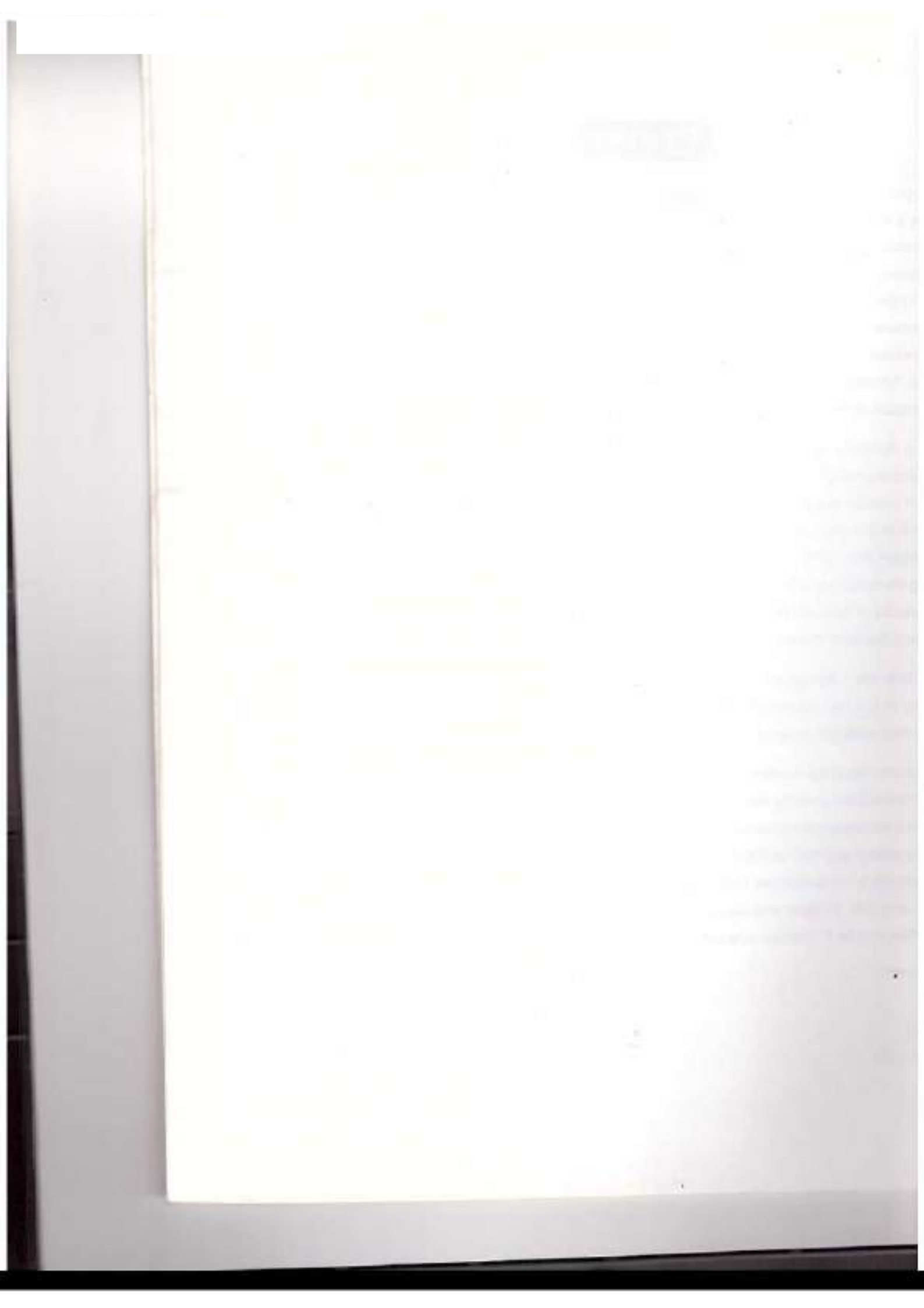
The Academy continued to extend policy support to the ICAR by developing policy documents viz. A Comprehensive Purchase Manual for ICAR; Constitution, Structure, Financial and Administrative Powers of Institute Management Committee; Human Resource Development Strategy for scientists, administrative and finance officers of ICAR. Capacity building through training in Strategic Research and Extension Plan (SREP) development activities was undertaken in collaboration with MANAGE in improving the bottom-up approach of agricultural research prioritization in the country. Keeping in view the importance of topicality of the subject, this year's Annual Report carries the theme paper entitled "Agricultural Research Prioritization Techniques for Preparation of Strategic Research Extension Plan".

A Web site - Agricultural Gateway of India with extensive links to important sites related to agricultural R & D was developed. As a recognition of its multifarious services during the year, the Academy was befittingly crowned with several awards and honours, both institutionally and individually.

Meteoric rise in the number of new initiatives would not have been possible but for the guidance and unstinted support given by Padma Bhushan Dr R.S. Paroda, Director General, ICAR, and Secretary DARE, Govt. of India and the professional support of Dr S.L. Mehta, Deputy Director General (Education). While cooperation and help of NAARM faculty and other functionaries were exemplary, I specifically owe a gratitude to my colleagues Drs S. N. Saha, T. Balaguru, D. Rama Rao, R. Kalpana Sastry, A. Gopalani, and B.S. Sontakki who supported in preparation of this report. Assistance of Mr R.V.V.S. Prakash Rao, and Mr P. Namdev in layout design and printing is highly appreciated.

Hyderabad
December 2001


J.C. Katyal
Director



1. Executive Summary

Ever since its inception in 1976, National Academy of Agricultural Research Management (NAARM) has played a pivotal role in strengthening the Indian National Agricultural Research System (NARS) through its multifarious contributions. It has been the lead centre in providing the human resource development backup to the NARS and policy support to the Indian Council of Agricultural Research (ICAR). During the year 2000-2001, NAARM continued to support the Indian NARS as well as those of the other developing countries in the areas of agricultural research and education management. Salient achievements of the institute are presented briefly.

As a part of the HRD activities, a wide variety of training programmes were organized by the Academy for the benefit of scientists / faculty from ICAR and State Agricultural Universities (SAUs) within the country as well as from the developing countries in Asia and Africa. During the reported period, 64 such programmes were held, and a total of 1158 scientists, teachers, administrators and finance officers participated in these programmes. The year witnessed a spurt in the international training programmes. From a total of five international human resource development programmes organized during the year, the academy trained 31 research managers from Sri Lanka, Tanzania and Yemen.

The Academy, through its research efforts, provides backup initiatives on improving the contribution and productivity of agricultural scientists of Indian NARS and those of other developing nations by superior management of agricultural research and education. On the research front, various institutional and externally funded projects were pursued actively. Important research achievements of the year include: future oil-seeds scenario in India through a NAARM developed technological forecasting methodology; development of a case study on institutionalization of micro-level prioritization of agriculture; revision of guidelines for preparation and implementation of Strategic Research Extension Plan (SREP) in collaboration with National Institute for Agricultural Extension Management (MANAGE), Hyderabad; development of basic framework for assessing the performance of agricultural research organizations towards ensuring their accountability; need assessment as a part of the development of HRD strategies for ICAR; identification of factors responsible for differential performance of NGO and GO, KVKs and development of case study on Integrated Farming Systems.

The institute continued to extend policy support to the ICAR by developing a comprehensive purchase manual for ICAR. Similarly, changes were suggested in the constitution, structure, financial and administrative powers of institute management committee to make it more effective. NAARM submitted a draft policy document on human resource development strategy for ICAR scientists and administrative and accounts officers.

During the year under report, the academy consolidated its information technology status. The NAARM web site was registered with ERNET with the URL <http://icar.naarm.ernet.in>. As many as 64 new internet protocol (IP)

Executive Summary

addresses were released by ERNET for use by various functionaries of NAARM. All faculty members of NAARM were provided independent IP addresses to enable reliable access to Internet through VSAT/ERNET. Faster internet access through 64 kbps ISDN lines were provided to faculty, staff and trainees for four hours on every working day. The NAARM web site was updated regularly and linked to the ICAR web site.

All these efforts and initiatives brought recognition to the institute in the form of awards to the members of the NAARM family. Two of its scientists won the second best article award among the articles published in the Fertilizer News. The Academy won the Rajashri Tandan Rajbhasha Puraskar instituted by the Indian Council of Agriculture Research, New Delhi for the year 1999-2000 for the best work done in the implementation of the Official Language in the 'C' region ('Ga' Kshetra). The NAARM Sports contingent, secured seven prizes in the finals of the ICAR Inter-zonal tournaments held at Central Institute of Fisheries Education, Mumbai. The Academy won a total of 15 prizes in the 15th Annual Rose Show conducted by the Horticultural Society, Secunderabad, and bagged the Hyderabad Rose Society Challenge Cup at the XIX All India Rose Show of Indian Rose Federation, and Silver Jubilee Rose Show of Hyderabad Rose Society.

2. Activities at a Glance

April 2000

- A ten-day programme on Statistical Software for Data Analysis was organized from April 4 to 15, 2000. Eleven scientists and technical officers from various ICAR institutes attended the programme. Drs K. Vidyasagar Rao and M. Narayana Reddy coordinated the programme.
- The Third Management Development Programme in Agricultural Research Management was organized from April 24 to 29, 2000. Eighteen Heads of the Departments and Project Coordinators of ICAR institutes participated in the programme. Drs Jagannadham Challa and S.N. Saha coordinated the programme.
- A Faculty Development Programme on Educational Psychology and Behavioural Technology was organized for the senior faculty members of Tamil Nadu Agricultural University, Coimbatore, from April 10 to 20, 2000. Eighteen faculty members attended the programme. Drs A. Gopalani, Jagannadham Challa, and R. Kalpana Sastry coordinated the programme.
- An Orientation Programme on Educational Administration and Management of Financial Issues was organized for the administrative and financial personnel of Tamil Nadu Agricultural University from April 29 to May 9, 2000. Twenty seven participants were benefited by attending this programme, which was coordinated by Drs K.P.C. Rao, A. Gopalani, and D. Rama Rao.
- A Sponsored programme on Computer Applications in Agricultural Research and Education was organized for the faculty members of Tamil Nadu Agricultural University, Coimbatore, from April 29 to May 4, 2000. Seventeen members attended the programme. Drs N. Hanumantha Rao, R.Chittiraichelvan, and Ajit P. Maru coordinated the programme.
- An International Programme on Agricultural Research Management was organized for Tanzanians from April 24 to June 2, 2000. Nine Tanzanians attended the programme, which was coordinated by Dr T. Balaguru.

May 2000

- A one-week Management Programme for women scientists was organized from May 1 to 6, 2000. Seventeen women scientists participated in the programme. Drs Janardan Jee and N. Sandhya Shenoy coordinated the programme.
- A two-day AHRD programme on Protected Agriculture was organized on May 10 and 11, 2000. Sixty scientists from SAUs, ICAR and staff of state departments and private sector attended the programme. Dr D. Rama Rao coordinated the programme. Prof. Medrle, H. Jensen, Arizona University and World Bank Consultant was the resource person.
- A one-week Faculty Development Programme on Educational Psychology and Behavioural Technology was organized exclusively for the teachers and senior level managers of Tamil Nadu Agricultural University, Coimbatore, from May 29 to June 3, 2000. Drs A. Gopalani, P. Manikandan, and P. Kumar coordinated the programme.

Activities at a Glance

- A ten-day programme on Improving Administrative Management in ICAR was organized from May 30 to June 9, 2000. Twenty six administrative and finance officers of ICAR institutes participated in the programme, which was coordinated by Dr S.N. Saha and Mr B. Pandu Reddy.

June 2000

- A ten-day programme on Computer Applications in Agriculture was organized from June 20 to July 1, 2000. Fourteen scientists and technical officers from ICAR and SAUs attended the programme. Drs K. Vidyasagar Rao and K.M. Reddy coordinated the programme.
- A ten-day programme on Rapid, Relaxed and Participatory Rural Appraisal and Participatory Learning and Action Techniques for Research and Extension in Agriculture was organized from June 20 to July 1, 2000. Scientists from ICAR institutes and SAUs, trainers from Krishi Vigyan Kendras and Farmers' Training Centres, numbering seventeen, participated in the programme. Drs VE. Sabarathnam and N. Sandhya Shenoy coordinated the programme.
- A ten-day Faculty Development Programme on Educational Psychology and Behavioural Technology was organized exclusively for the senior faculty members of Tamil Nadu Agricultural University, Coimbatore from June 26 to July 6, 2000. Twenty two members were benefited by participation in this programme. Drs A. Gopalani, P. Manikandan, and S. Shanmugam coordinated this sponsored programme.

July 2000

- A five-day Workshop-cum-Training on Official Language Policy for Local ICAR Institutes was organized from July 3 to 7, 2000. Thirty eight officials of various local ICAR institutes participated in the workshop. Dr A. Gopalani and Mr S. Pradeep Singh coordinated the workshop.
- The Third Executive Development Programme in Agricultural Research Management was organized from July 15 to 18, 2000. Twelve Heads of the Departments and Directors of various ICAR institutes participated in the programme. Drs S.N. Saha and Jagannadham Challa coordinated the programme.
- A programme on Agricultural Scientist Development for Personal and Organizational Effectiveness was organized from July 17 to 28, 2000. Eleven scientists from ICAR institutes participated in the programme. Drs S. Shanmugam and Jagannadham Challa coordinated the programme.
- A two-week programme on Educational Video Production was organized from July 18 to 29, 2000. Four scientists and technical officers of ICAR institutes participated in the programme. Dr R.C. Chelvan and Mr K.R. Prabhakar coordinated the programme.
- A two-week programme on Instructional Aids for Effective Presentation was organized from July 18 to 29, 2000. Two members attended the programme. Drs N. Sandhya Shenoy and Janardan Jee coordinated the programme.
- A one-week programme on Computer Applications in Agricultural Research and Education was organized for Tamil Nadu Agricultural University faculty members from July 17 to 22, 2000. Fifteen faculty members attended the programme. Drs N. Hanumantha Rao, K. Vidyasagar Rao, and R.C. Chelvan coordinated the programme.

- An orientation programme on Educational Administration and Management of Financial Issues was organized for the functionaries of Administrative and Finance wings of Tamil Nadu Agricultural University from July 17 to 27, 2000. Twenty seven members attended the programme. Drs K.P.C. Rao, Janardan Jee, and A. Gopalam coordinated the programme.

August 2000

- A ten-day programme on Analytical Tools and Techniques for Assessing Agricultural Sustainability was organized from August 1 to 10, 2000. Seven scientists participated in the programme. Drs N. Hanumantha Rao and M. Narayana Reddy coordinated the programme.
- A one-week programme on Agricultural Research Prioritization Techniques was organized from August 21 to 26, 2000. Thirty one members participated in the programme. Drs K.P.C. Rao and B.S. Chandel coordinated the programme.

September 2000

- A one-week programme on Participatory Methods in Agricultural R & D was organized from September 11 to 16, 2000. Seven scientists from various ICAR institutes participated in the programme. Drs N. Sandhya Shenoy and VE. Sabarathnam coordinated the programme.
- A one-week programme on Learning to Lead and Succeed in New Environment was organized from September 14 to 20, 2000. Four scientists from ICAR institutes participated in the programme. Drs Janardan Jee and Jagannadham Challa coordinated of the programme.
- A programme on Computer Awareness was organized from September 18 to 23, 2000. Nineteen scientists and technical officers of ICAR institutes participated in the programme. Dr K.M. Reddy coordinated the programme.
- A ten-day programme on Video Editing Techniques was organized from September 19 to 29, 2000. Six technical personnel engaged in videography in various research institutes participated in the programme. Mr K.R. Prabhakar coordinated the programme.
- A one-week programme on Impact Assessment of Agricultural R & D was organized from September 25 to 30, 2000. Fifteen scientists of ICAR institutes and SAUs who are having knowledge on various mathematical econometric techniques participated in the programme. Drs B.S. Chandel and K.P.C. Rao coordinated the programme.
- A programme on Decision Support Systems was organized from September 25 to October 15, 2000. Twenty one scientists and faculty members from various ICAR institutes and SAUs participated in the programme. Drs D. Rama Rao and S.K. Nanda coordinated the programme.

Activities at a Glance

- A ten-day Faculty Development Programme on Educational Psychology and Behavioural Technology was organized exclusively for the teachers and senior level managers of Tamil Nadu Agricultural University, Coimbatore from September 11 to 21, 2000. Dr T. Balaguru coordinated the programme.
- A three-day training-cum-workshop on PIMs was organized from September 27 to 29, 2000, by IASRI. Scientists participated in the programme. Dr M. Narayana Reddy coordinated the programme.

October 2000

- A one-week programme on Project Management Using Microsoft Project was organized from October 30 to November 4, 2000. Fifteen members attended the programme. Dr D. Rama Rao coordinated the programme.

November 2000

- An International Programme on Agricultural Research Management was organized for two Directors of Agriculture, Republic of Yemen, from November 1 to December 30, 2000. Dr T. Balaguru coordinated the programme.
- A ten-day programme on Geographical Information System Applications in Agriculture was organized from November 6 to 16, 2000. Thirteen members attended the programme. Drs N. Hanumantha Rao and M. Narayana Reddy coordinated the programme.
- Reorientation Programme on Educational Administration and Management of Financial Issues was organized for senior level managers of Tamil Nadu Agricultural University, Coimbatore from November 7 to 17, 2000. Twenty six members attended the programme. Dr A. Gopalani, Capt. Mehar Singh, and Mr B. Pandu Reddy coordinated the programme.
- A ten-day programme on Agricultural Research Project Management was organized from November 14 to 24, 2000. Eleven members participated in the programme. Drs S.K. Nanda and D. Rama Rao coordinated the programme.
- Two training programmes on Computer Applications for Agricultural Research and Education for Tamil Nadu Agricultural University staff were organized from November 13 to 18, 2000 and from November 18 to 23, 2000. A total of 30 members attended the programme. The first programme was coordinated by Drs K. Vidyasagar Rao, N. Hanumantha Rao, M. Narayana Reddy, and the second programme by Drs D. Rama Rao, N. Hanumantha Rao, and R. Kalpana Sastry.
- A one-week programme on Indian Seed Industry Management was organized from November 20 to 25, 2000. Twenty two members participated in the programme, coordinated by Drs K.P.C. Rao and Janardan Jee.
- A Workshop on Official Language Policy Resolution of New Millennium was organized on November 28 and 29, 2000. Forty seven members attended the workshop. Dr A. Gopalani and Shri D. Venkateswarlu coordinated the workshop.

- A ten-day course on **Advanced Data Analytical Techniques for Research and Management** was organized from November 28 to December 8, 2000. Eight members attended the course. Drs M. Narayana Reddy and K. Vidyasagar Rao coordinated the programme.
- Fourth Management Development Programme in Agricultural Research was organized from November 30 to December 6, 2000. Thirteen Heads of the Divisions of ICAR institutes participated in the programme. Drs Jagannadham Challa and S.N. Saha coordinated the programme.

December 2000

- One-day **Brainstorming Session on Good Governance in Agricultural Research Organization** was organized on December 2, 2000. Nineteen fellows of National Academy of Agricultural Sciences and faculty members of NAARM attended the session. Drs J.C. Katyal and N. Hanumantha Rao coordinated the session.
- A ten-day programme on **Improving Administrative and Financial Management in ICAR institutes** was organized from December 6 to 16, 2000. Seventeen administrative and finance officers from various ICAR institutes participated in the programme. Dr S.N. Saha, Capt. Mehar Singh and Mr B. Pandu Reddy coordinated the programme.
- A two-week programme on **Computer Applications in Agriculture** was organized from December 12 to 23, 2000. Nine scientists from various ICAR institutes attended the programme. Drs K.M. Reddy and K. Vidyasagar Rao coordinated the programme.
- A two-week programme on **Educational Video Production** was organized from December 12 to 23, 2000. Three technical officers from various ICAR institutes attended the programme. Mr K.R. Prabhakar and Dr Janardan Jee coordinated the programme.
- A two-day **National Conference on System Dynamics for Agricultural Development** was organized on December 28 and 29, 2000 in collaboration with Systems Dynamics Society of India (SDSI). A pre-conference training on System Dynamics was organized on December 26 and 27, 2000. Forty delegates including scientists and officials from ICAR, Indian Institutes of Technology, Tata Consultancy Services, MANAGE, SAUs and other organizations participated in the conference, which was coordinated by Dr D. Rama Rao.

January 2001

- A three-day training on **Designing and Developing Web Pages for the Scientists of Agricultural Technology Information Centres (ATIC's)** was organized from January 17 to 19, 2001. Twenty scientists from ATICs attended the programme. Drs N. Sandhya Shenoy and M.N. Reddy coordinated the programme.
- A one-week course (under NATP) on **Agricultural Research Prioritization Techniques** was organized from January 18 to 24, 2001. Fifteen members attended the course. Drs K.P.C. Rao and B.S. Chandel were the coordinators.
- A workshop on **Gahan Hindi Prashikshan** was organized from January 22 to 27, 2001. Forty three members attended the workshop. Dr A. Gopalam and Mrs J. Renuka coordinated the workshop.

Activities at a Glance

- An International Programme for Administrators and Accountants from Sri Lanka was organized from January 22 to 27, 2001. Seven Administrators and Accountants from Sri Lanka attended the programme. Drs T. Balakrishnan, R. Kalpana Sastry and M.N. Reddy coordinated the programme.

February 2001

- An Executive Development Programme on Agricultural Research Management was organized from February 1 to 5, 2001. Six Heads of the Departments and Directors of various ICAR institutes attended the programme. Drs S.N. Saha and Jagannadham Challa coordinated the programme.
- A three-day workshop on Leadership and Management for Tamil Nadu Agricultural University Deans and Directors was organized from February 8 to 10, 2001. Twenty two members attended the workshop, coordinated by A. Gopalam and P. Manikandan.
- A ten-day course on Computer Applications in Agriculture was organized from February 13 to 23, 2001. Twenty scientists and technical officers from ICAR institutes participated in the programme. Drs K. Vidyasagar and K.M. Reddy coordinated the programme.
- A two-week International Programme on Agricultural Research Management for Sri Lankan Scientists was organized from February 22 to March 7, 2001. Three scientists from Sri Lanka attended the programme. S.N. Saha, S. Shanmugam, and S.K. Nanda coordinated the programme.

March 2001

- A three-week International Programme on Agricultural Research Management for Sri Lankan scientists was organized from March 12 to 31, 2001. Ten scientists from Sri Lanka participated in the programme. Rama Rao, Janardan Jee, and B.S. Chandel coordinated the programme.
- A one-week course on Techniques for Effective Linkages in Agriculture was organized from March 19 to 24, 2001. Seven scientists from different ICAR institutes participated in the programme. Drs V.E. Sabarwal, V.K.J.R. Rao, and B.S. Sontakki coordinated the course.
- A one-week Management Programme for Women Scientists was organized from March 19 to 24, 2001. Three women scientists participated in the programme. Drs Janardan Jee and N. Sandhya Shenoy coordinated the programme.
- A ten-day course on Statistical Software for Data Analysis was organized from March 20 to 31, 2001. Twenty scientists and technical officers from ICAR institutes attended the course. Drs K. Vidyasagar Rao and Narayana Reddy coordinated the course.
- A one-week Computer-based Instructional Technology programme was organized for the faculty members of Tamil Nadu Agricultural University from March 2 to 9, 2001. Twenty four faculty members from different departments of TNAU participated in the programme. Drs A. Gopalam, N. Hanumantha Rao, and M. Narayanan coordinated the programme.

Agricultural Research Prioritization Techniques for Preparation of Strategic Research Extension Plan

Dr K.P.C. Rao, Principal Scientist

Agricultural research prioritization refers to the process of ranking a set of potential research activities so that the scarce research resources may be allocated to these activities as per the ranking. This process is undertaken to align the resource allocation with the national objectives and to improve the effectiveness and the efficiency of the scarce research resources. It also aims to improve the transparency and accountability in the research systems with respect to resource allocation, monitoring and evaluation.

3.1 The Need and Rationale of Research Prioritization

The relevance of research prioritization rests on the following assumptions:

- ❶ There is always a higher demand for research resources like scientific and technical manpower, capital assets and contingencies than their availability. If the supply of research resources matches or exceeds the demand for them, the research prioritization exercises would not be relevant.
- ❷ Systematic and objective approaches lead to a more efficient resource allocation than the informal mechanisms such as collective judgment or beliefs of individual scientists and/or research managers.
- ❸ A good priority-setting approach will allow information on national policy objectives to flow downward, and information on researchable problems to flow from the bottom up to influence higher levels of priority setting.
- ❹ Institutionalization of priority assessment mechanisms at different levels would enhance transparency, ownership and accountability, thereby contributing to the effectiveness and efficiency of the research system.

3.2 Apprehensions about Research Prioritization

Research prioritization being a relatively new concept, there are apprehensions among some scientists and research managers about it. The main arguments against research prioritization are summarized below:

- ❶ Research is an uncertain economic activity with unpredictable outcomes. Any pre-judgement of research activities is not fair. If scientists have to convince their peers or managers about the desirability of the research activities they wish to take up, they may tend to shift from basic and upstream research to applied and downstream research.
- ❷ The informal mechanisms used for research resource allocation in the past have been quite effective in meeting the national priorities.
- ❸ The best research prioritization approaches cannot guarantee a close correspondence between the *ex-ante* assessment and *ex-post* impact of any research activity.
- ❹ The methodologies suggested by economists and other social scientists are only a ploy to take away the decision making powers and prerogatives of the research managers.

Despite such apprehensions and objections voiced by some scientists and research managers, the priority assessment procedures are finding increasing application and acceptance in the western countries. The National

Agricultural Research System (NARS) has decided to make use of the funding under National Agricultural Technology Project (NATP) for introducing Organizational and Management (O&M) reforms to improve the transparency, accountability and efficiency of the system. The processes of priority assessment, monitoring and evaluation (PME) are sought to be strengthened through sensitization, capacity building and institutionalization of the necessary procedures. Priority assessment at different levels of research planning is the first step in ushering the O&M reforms in the NARS.

3.3 Levels of Research Resource Allocation

Although research resources are allocated at several levels, three main levels can be identified:

- i) Allocation at the macro-level, especially allocation across states (or regions), commodities and resource management programmes within a national research system.
- ii) Allocation at the programme or state levels, such as the share of resources going to different lines of work under a commodity like crop improvement, crop management, crop protection etc., or the allocation done at a state or university level across regions or research stations.
- iii) Allocation at the level of institutes or regional research stations across projects for carrying out specific time-bound experimental programmes.

But the three levels referred above are not mutually exclusive. The NARS has central research institutes and national research centres dealing with a single commodity or with a group of commodities but catering to the needs of a wider geographical area. Although these institutes have a wide geographical mandate, their research agenda gets influenced by areas more proximate to it than the areas farther from it. Commodity orientation has been quite strong even in the SAUs, with a particular research station spearheading the research on a particular commodity and catering to the needs of the whole state. But after the implementation of the National Agricultural Research Project (NARP), the regional focus is emphasized more than the commodity focus in them. Thus, there can be a wide variety of situations for priority assessment. A national level institute may have the simple task of allocating resources between sub-programmes and projects dealing with a single commodity. At the other end, a regional agricultural research station may be attempting resource allocation between projects dealing with several commodities and resource management aspects. There are also the issues of technology distance, spill-in and spill-over possibilities of technologies and the desired level of duplication of similar research work at more than one place. Institutionalization of the process of priority assessment at different levels of NARS is a challenging task.

3.4 Research Philosophy: Supply-determined or demand-driven?

The NARS is funded by both the union and state governments. The Department of Agricultural Research and Education (DARE) provides the link between the Union Government and the Indian Council of Agricultural Research (ICAR). The ICAR, besides running its own institutes, research centres, project directorates and coordinated projects, also funds education and research activities in the SAUs. The state governments provide the bulk of funding for the state agricultural universities, while the funding from ICAR also makes a substantial contribution to them. Because the funds flow from the top to the bottom, the research agenda also gets influenced by the perceptions and priorities of the research managers at the top levels of these organizations. While this may be necessary to give importance to the national priorities and objectives, there is a danger that the research agenda may not be responsive to the changing needs and priorities of the stakeholders, mainly the farmers. Although there is a need for the researchers

to keep abreast of the technological breakthroughs being achieved in different parts of the World in the frontier areas of science and technology so as to make use of them in their research programmes and find ways and means of reducing the unit costs of production, there is also a need to respond to location-specific problems which are constraining agricultural production and farmers' income. Yet, the research agenda gets influenced by the opinions of the higher-ups and peers, rather than by the needs of the stakeholders. The technologies that come out of the research stations are generally supply-determined rather than demand-driven. One classic example is that of the agro-forestry systems recommended by different agro-forestry research centres. These systems generally have either eucalyptus or subabul as the base tree species. But these systems have little or zero acceptability of the farmers because the farmers may require timber or fodder or some other products rather than the firewood, which eucalyptus or subabul can give. Eucalyptus and / or Subabul are considered quick growing along with some other desirable traits and are, hence, included in the agro-forestry systems. Selection of such tree species is an example of supply-orientation without bothering about catering to the needs of the farmers. Although there is enough appreciation of both the top-down as well as bottom-up approaches for determining the research agenda, it is the top-down approach which is mostly followed in practice. The recommended technologies are, sometimes, found to be out of tune with the resource endowments and preferences of the farmers and, hence, are adopted by the farmers to a limited extent. Votaries of 'farmer-first and farmer-last' approach like Robert Chambers emphasized the need for a bottom up approach. A research agenda that emerges in response to the problems and priorities of the stakeholders is likely to be more fruitful. The research prioritization techniques that can be used for prioritization of research programmes and projects differ with the top-down or bottom-up approaches used in generating research agenda. Certain techniques like congruence method or Domestic Resource Cost Ratio (DRCR) methods are generally used at a macro-level (at the level of a country or a state). Other techniques like checklist methods, scoring models and benefit – cost ratio analysis are used for prioritization both at the macro as well as at the micro-levels, irrespective of the top-down or bottom-up approach used for generation of the research projects.

3.5 Regional Research and Bottom-up Approaches

There is a growing realization among the policy makers that there has been an excessive emphasis on attaining a fast growth in the production of agricultural commodities to the detriment and neglect of natural resources over the last five decades. The research managers are worried about resource degradation in the face of which the future growth appears uncertain and unsustainable. The need to improve the quality and productivity of the main agricultural resources, land and water, calls for a shift from a commodity-focus to a regional focus. Regional research aims to optimize current agricultural production and natural resource management systems.

The NARP gave a fillip to the regional research by supporting the infrastructure development at 126 regional agricultural research stations in the country. For each of these regions, a status report was prepared. The status report typically delineates the various farming situations within the agro-climatic region and surveys the production constraints and research problems, based on which the scientists were expected to develop research projects. A Zonal Research and Extension Advisory Committee (ZREAC) was set up with representatives from the farmers and the extension departments figuring in it. It was done mainly to strengthen the feedback mechanism. In general, the NARP laid the foundations of regional research and bottom-up feedback processes.

In the earlier stages of National Agricultural Technology Project (NATP) formulation, the regional research focus was further emphasized. But subsequently, the regional focus was considerably watered down in the final formulation. Although the production system research received about 58 percent of the total allocation under the project, the delineation of these systems entailed a compromise between the regional focus and commodity orientation.

The first level delineation into five agro-eco systems: the coastal, the hill and sub-montane, the arid, the rainfed and the irrigated eco-systems was based on geo-climatic factors. But the next level of delineation was based on the commodities. For instance, the rainfed agro-ecosystem was classified into eight sub-systems based on the main commodity grown viz., rice, sorghum, pearl millet, finger millet, cotton, soybean, groundnut and rapeseed and mustard. The dilution of the regional focus has also led to the top-down approach in the generation of the research agenda. The lead and cooperating centres as well as the principal and co-investigators were picked up by the top research managers for some of the projects, atleast initially. Prioritization exercises were carried out much later than the approval of the projects. Even the subsequent research proposals submitted by the scientists were not based on any bottom-up exercises. In several cases, the on-going work was repackaged into new projects. The National Centre for Agricultural Economics and Policy Research (NCAP) took up the task of identification and prioritization of production constraints in rainfed and irrigated agro-eco systems after the allocation of funds to different projects was completed in these agro-ecosystems. NAARM has undertaken a similar task in the coastal agro-ecosystem.

3.6 Research Resource Allocation and Research Prioritization

Both these processes are interlinked. In every research resource allocation, there may be an implicit prioritization, whether it is made explicit or not. In general, a method of incrementalism is followed in allocating resources in the government departments. If the research organization gets a ten per cent higher allocation in the current year over that of the previous year, the tendency is to enhance the allocations to different institutes or research programmes by the same ten per cent. However, some marginal variations can be made here and there as exceptions. But this method of allocation places immense faith on the past resource allocation decisions. Another reason for this method of allocation is that the normal increases in the committed expenditures like salaries, POL, labour charges etc. are to be taken care of.

3.6.1 Congruence Method

There has been a criticism of such an arbitrary method of resource allocation in the system. An objective method of resource allocation which follows a rule of thumb has been developed and christened as the 'congruence method'. This method suggests that the research funds must be allocated to commodities or regions in the same proportion as their existing contribution to the Gross Domestic Product emanating from the agricultural sector. This method ensures certain minimum allocation of research resources to each agricultural commodity/region. But its drawbacks are that, it is not useful for non-commodity research and that it favours the already well established commodities and ignores the new ones. The implicit assumption behind it is that opportunities for research are equal across commodities and regions and that the value of new knowledge generated by research is proportional to the value of output. It also assumes that the relative shares of different commodities will be constant. Despite these restrictive assumptions, this method provides an objective, transparent and rational basis for allocation of research resources. The Technical Advisory Committee (TAC) of the Consultative Group on International Agricultural Research (CGIAR) also opted for this approach because of its simplicity, transparency and flexibility in its study on priorities for international agricultural research.

3.6.2 Domestic Resource Cost Ratio Method

After the signing of the World Trade Agreement (WTA) by a large majority of the countries in the world, the export competitiveness has become an important issue in the face of impending globalization of trade in agricultural commodities. If the countries of the world wish to take advantage of the trading opportunities, they have to give a

higher priority for the commodities in which they enjoy a comparative advantage. The computation of the Domestic Resource Cost Ratios (DRCRs) and using them as indicators of competitiveness for determining the priority and altering the resource allocation due to them may gain importance in the future.

$$DRCR = A / (B-C)$$

Where 'A' is the unit cost of producing a commodity domestically
 'B' is the unit cost of importing a commodity from other countries
 'C' is the cost of imported inputs used in domestic production (per unit output)

When this ratio is less than one, that commodity has to be accorded a higher priority than what is warranted as per the congruence analysis. If the ratio is more than one, the commodity has to receive a relatively lower priority. But, there are serious data limitations to work out these Domestic Resource Cost Ratios, as there are no estimates of imported inputs used in the production of different commodities. In the future, as these data gaps get filled up, this method may also be employed more widely to determine the allocation of research resources.

These two methods, Congruence and Domestic Resource Cost Ratios, are useful for resource allocation at the level of a country or a state. The first level allocation may be done between commodity research and non-commodity research using 75:25 or 60:40 or 50:50 or any other ratio. The funds earmarked for commodity research may further be apportioned between different commodities based on either Congruence or Domestic Resource Cost Ratio method. The funds earmarked for non-commodity research may be allocated further to resource management research, genetic conservation research, social science research etc., following some other method or criterion.

3.6.3 A Critique on Checklist and Scoring Methods

The implicit rules of prioritization in the Congruence and DRCR methods discussed above are contribution to agricultural GDP and comparative advantage respectively. Yet, they are considered more as resource allocation methods than as the methods of research prioritization.

The checklist and scoring models are more explicit methods of research prioritization. These methods are more versatile in application as they can be used for prioritization and resource allocation for commodities, individual lines of work under a commodity, non-commodity research projects as well as for social science research projects. The checklist method basically lists a set of criteria requiring answers to set priorities for research programmes and projects. Depending on the degree of the satisfaction of the set criteria, the programmes and projects are ranked. Because of its simplicity and non-mathematical approach, checklist method is very much liked by the agro-biological scientists. But its limitation is that it does not indicate the relative significance of one criterion against the other.

A more sophisticated version of the checklist method in which relative weights are assigned to the criteria chosen for research prioritization is known as the scoring or weighted criterion method. Besides assigning relative weights to different chosen criteria, scores are also assigned to each project based on the degree of satisfaction with respect to each criterion. A weighted score is computed for each research project based on which the priority to be assigned is determined. This method allows for collection of opinions of several experts about the relative weights to be assigned to different criteria to make it a more transparent and participatory process. Yet, an element of subjectivity still remains in the assignment of scores and weights, although its magnitude is minimized and averaged.

3.6.4 Benefit – Cost (Economic Surplus) Approach

A typical research programme or project requires investments to be made initially to execute them. There is some uncertainty associated with the accomplishment of objectives. Even if the research objectives are achieved, the technologies generated may be adopted by only a certain proportion of the farmers and the full impact intended may not be realized. The impact of a research programme or project may be felt only after a considerable length of time period. Because of the long time lags involved between investments and returns, discounting of costs and benefits to a common point of time is necessary for any meaningful comparison. The economic surplus approach estimates returns to investments by estimating the benefits from research in terms of the change in consumer and producer surpluses that result from technological change. *Ex-ante* analysis usually incorporates expert opinions to determine projected research impacts, adoption rates and probabilities of research success and provide estimates of the economic efficiency and distribution implications of agricultural research resource allocation. The typical effect of technological change is to cause a rightward shift in the supply curve due to a reduction in the unit cost of production. This results in a new equilibrium price, which is lower than the earlier price and a new equilibrium quantity, which is higher than the earlier quantity. Consumers gain from the adoption of the new technology because they can consume more at a lower price, and the producers gain because their unit production costs have fallen. Net social benefit is the sum of consumer and producer surplus. The shares of the consumers and producers in the total benefits depend on the elasticities of demand and supply curves and the size of the supply shift. This economic surplus method is used to examine the general equilibrium effects of research, to assess the spill over of research benefits among different technologies, commodities, regions or countries, and to estimate the effects of agricultural policies or benefits arising from research. However, it is difficult to apply it to a large number of commodities or research areas because of data limitations and computational problems. The agro-biological scientists do not like this method as it calls for the involvement of economists or other social scientists. But, International donor agencies generally insist up on a benefit-cost analysis in an *ex ante* sense before taking a final decision to fund the projects. As the research programmes and projects are akin to any other types of developmental projects, it is imperative that they are subjected to benefit-cost analysis by discounting them at an appropriate rate of discount. After the necessary computations are done, programmes or projects may be prioritized based on the values of the net benefit-cost ratios or the net present values or the internal rates of return.

3.7 Prioritization of Production Constraints and Problems

We have discussed above about some techniques of agricultural research prioritization, which can provide some guidance for allocation of research resources when they are scarce and limiting. When we follow the bottom-up approach for the development of the research agenda, there is a need to identify the researchable problems and when their number is too large to receive the immediate attention of the researchers, it is also necessary to prioritise the researchable problems, so that the most important of them receive the immediate attention of the researchers. In a bottom-up process of research programme development, the first level prioritisation is done at the problems level and second level prioritisation is done at the research projects stage. There can be several approaches to the identification and prioritisation of the researchable problems.

The National Centre for Agricultural Economics and Policy Research (NCAP) followed a survey method for identifying the client constraints and research needs. An exhaustive list of constraints for major crops as well as livestock was prepared by reviewing the literature and through interactions with the extension workers and scientists. Subsequently, the constraints having economic significance were short-listed and detailed information was gathered on them from a sample of farmers. The severity of each constraint was estimated in terms of the yield lost due to that

constraint. The farmers were also asked to estimate the proportion of area affected by that constraint out of the total area under the crop in that village. They were also asked to estimate the long term probability of occurrence of that constraint in that village on the basis of their past experience. These estimates of percent yield loss, percent area affected and the probability of occurrence were then cross-checked and made more reliable by using the long experience, accumulated knowledge and pragmatic judgment of scientists and extension personnel working in that region. By multiplying the estimates of percent yield loss, percent area affected and probability of occurrence, the average yield loss due to the constraint was estimated. This average yield loss multiplied by the prices of concerned crop (MSP recommended by the Commission on Agricultural Costs and Prices) provided the estimated value of production foregone. The total losses from technical constraints were subtracted from the total yield gap and the residual was attributed to the socio-economic constraints.

In this method, yield gap is obtained as the difference between the research station yield and the actual average yield of the farmers. It is necessary that the research station yield and the average yield of the farmers should be drawn from the same season to take care of all other differences. But, it is not uncommon to pick up the maximum research station yield recorded so far and the average yield of the farmers recorded in some other season. This approach also assumes implicitly that the research stations have already solved all the researchable problems.

But, it is not necessary to make this restrictive assumption. For the prioritization of problems, it is enough to establish and quantify the relative importance of different researchable problems. An alternative approach used in the preparation of Strategic Research and Extension Plans (NAARM - MANAGE Collaborative work) deserves to be mentioned.

3.8 What is SREP?

The Strategic Research Extension Plan (SREP) is the plan to be implemented by the Agricultural Technology Management Agency (ATMA) in each of the 24 districts belonging to six states of Andhra Pradesh, Orissa, Bihar, Maharashtra, Punjab and Himachal Pradesh over the period of implementation under the NATP. The important focus of the exercise in the development of SREP is to improve the farmers' input into programme planning by involving them in participatory appraisals. A thorough analysis of the feedback so obtained helps in developing the strategies and activities based on the research and extension gaps identified for the districts. In order to do this, the ATMA is responsible for the proper planning and implementation of this exercise with the help of National Institute of Agricultural Extension Management (MANAGE) / National Academy of Agricultural Research Management (NAARM) / State Agricultural Management and Extension Training Institute (SAMETI). It is a participatory methodology to prepare strategic research extension plan at the district level to increase agricultural production; to formulate research and extension agenda based on producers' requirement; to develop technology acceptable to users and to prioritise resource allocation to research and extension activities at the district level.

Better farmer's organization and identification of technical gaps that lead to better natural resource management, if implemented effectively and efficiently can enhance the farming system focus.

3.8.1 Present Weaknesses in the Technology Development and Transfer

Presently the existing research and extension framework is top down, and scientists decide research priorities rather than the farmers influencing them. Since scientists have limited exposure to field realities and the involvement of extension and farmers is passive, the present feed back system is weak. Further, the technology recommendations

are too general and are not suitable to the multiple farming situations available within a district. Therefore, refinement of technological packages for different farming situations is needed. Before refining the technologies, the research and extension gaps need to be identified and prioritized. The SREP is a tool to achieve this objective.

3.8.2 Why Priority Setting?

The primary objective of priority setting is to allocate scarce research resources efficiently i.e. to maximize the benefit to the society from public expenditure for research. The Innovations in Technology Dissemination (ITD) component of NATP attempts to identify and prioritize researchable issues. The prioritization of researchable problems was carried out in Chittoor and Adilabad districts of Andhra Pradesh with the following objectives.

- i. To select best portfolio of research activities.
- ii. To review the existing resource allocation.
- iii. To compare existing and new alternatives.

3.8.3 Approach used for Chittoor and Adilabad SREPs

In the SREP exercises, the researchable problems were identified through the Participatory Rural Appraisals carried out in a set of villages representing a particular agro-ecological situation. In order to establish their relative importance, each member of the Team of Farm Advisors (TOFA) was interviewed to obtain their estimates of per cent losses (ranging between 0 and 1) caused by different problems, the class of farmers typically affected by it (giving a score of 1 for large farmers, 2 for medium farmers and 3 for small farmers) and the urgency of the problem (giving a score of 1 to least urgent and 10 to the most urgent). In order to obtain the magnitude of the problem, the above three values were multiplied for each of the researchable problems. Then, the weighted scores given by different team members were averaged. The researchable problems were then ranked by the average values in the descending order. Thus, a list of prioritized problems in a given agro-ecological situation was obtained. The researchable problems which are common to the entire district were prioritized by further averaging the values across agro-eco situations. Then, a list of prioritized researchable problems for the district as a whole was arrived at. Along with the prioritized researchable problems for the entire district, lists of researchable problems which are unique to different agro-ecological situations were also prepared.

The above simple methodology provided a quantitative basis for the identification and prioritization of researchable problems at the agro-ecological situation level as well as at the level of the district. The ATMA authorities can consider funding research stations within and around the district to take up research projects for solving researchable problems that are unique to the district in question. Some research problems which may be of interest to a much wider geographical area can be referred by ATMA to the nearby regional research station of the state agricultural university or the relevant ICAR institute, particularly when basic research is needed for their redressal. In any case, arriving at a list of researchable problems is the pre-requisite for taking necessary action by the ATMA authorities. The exercises carried out for the preparation of SREPs in Chittoor and Adilabad districts of Andhra Pradesh have demonstrated the utility of the above systematic approach to prioritize the researchable problems. MANAGE has undertaken to spread this methodology to other SREP districts where preparation of SREPs is progressing by formally amending the SREP guidelines suitably. NAARM, on its part, is extending to MANAGE and ATMAs in different parts of the country in implementing this changed methodology. NAARM-MANAGE collaboration in improving the methodology of SREP preparation is a significant contribution to the bottom-up approach of agricultural research prioritization in the country.

4. Introduction

The National Academy of Agricultural Research Management (NAARM), one of the five National Institutes of the Indian Council of Agricultural Research (ICAR), New Delhi, was established in 1976, to fulfill the important need for an institution of training of agricultural scientists in the broad areas of management of agricultural research and education.

Apart from its quantitative contributions in the areas of training, research, policy support and consultancy, the academy is contributing significantly to generate a sense of fraternity and inculcate scientific culture among the agricultural scientists in the country. This unique institution has already established its credibility in the national and international arena. It is looked upon as a model institute by the developing countries and it is fast evolving into a viable Regional Training Centre in this part of the world to train agricultural scientists in agricultural research and education management.

Over the years, the Academy with support from ICAR has been able to build up an excellent infrastructure, state-of-the-art video production laboratory, agricultural information unit and other facilities necessary for imparting training of highest quality. The Academy is endowed with an excellent Library and Information Centre, Conference Hall, Seminar Hall, Committee Room, Scientist Home, Halls of Residence, Computer lab, Offset Printing Press and Health Centre.

4.1 Mandate

NAARM is charged to fulfil the following mandate :

- To organize and conduct training programmes in agricultural research management for the scientists at various levels
- To build up high quality resource material in agricultural research management based on actual field experience
- To undertake systematic review and study of management problems of agricultural research institutes, programmes and systems
- To plan, organize and conduct workshops and seminars in research management and educational technology
- To organize, liaise and coordinate programmes of international cooperation in the field of agricultural research management
- To assist the State Agricultural Universities in developing Regional Centres of Management to cater to the needs of various Universities

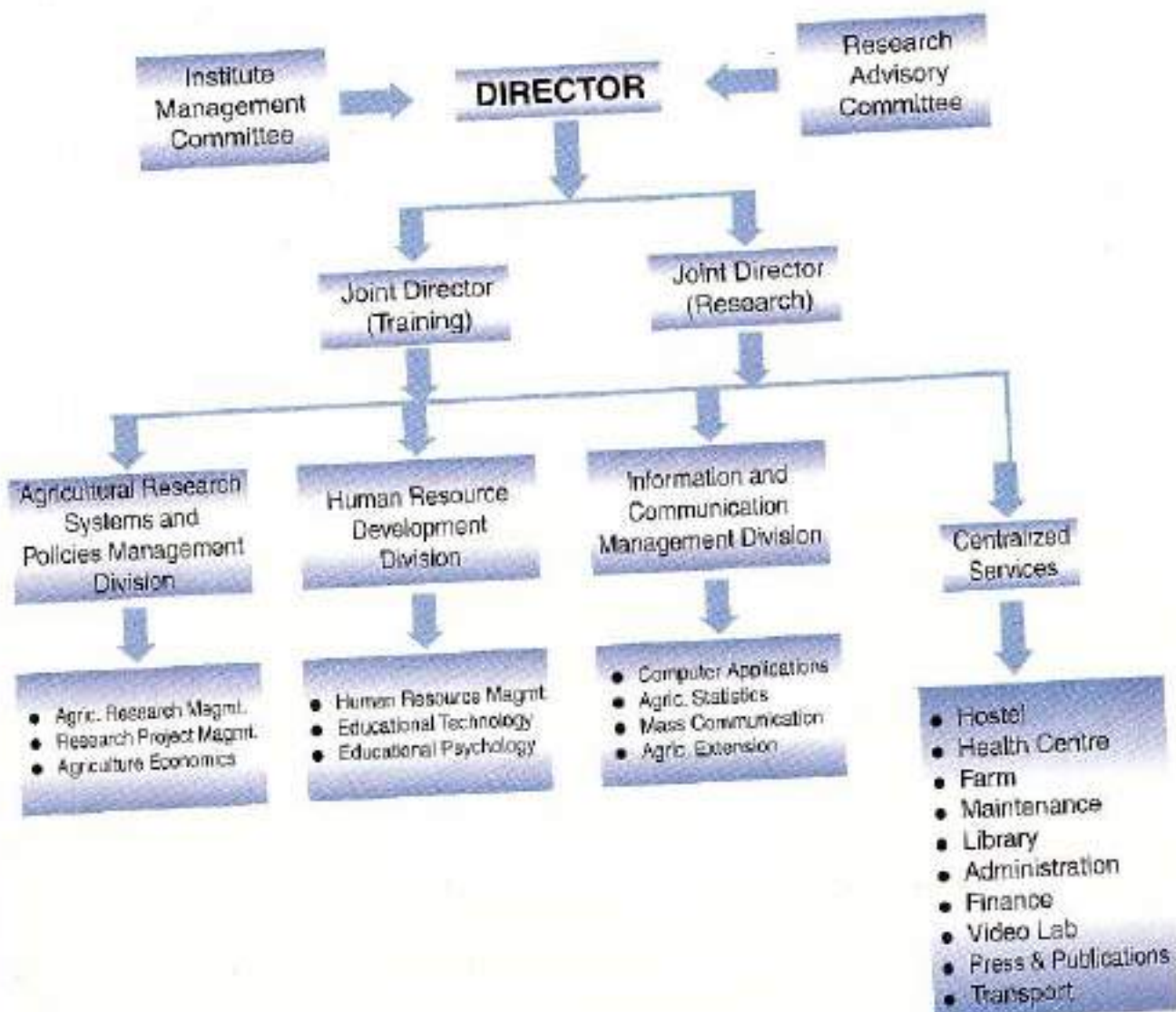
4.2 Objectives

- To organize foundation courses for the scientists newly recruited into the Agricultural Research Service of ICAR
- To design and organize training programmes, workshops, and seminars, and to develop high quality material in agricultural research and education management for the senior and middle-level scientists of NARS
- To play a major role in different processes of human resource development of NARS, in the context of agricultural research and education
- To undertake policy-level studies on prioritization of research and resource allocation, and to enhance research productivity of the constituents of NARS and their programmes
- To evolve suitable management techniques for effective planning, scheduling, organizing, monitoring, and evaluation of agricultural research and development projects in NARS
- To study the management issues of various programmes and patterns of technology transfer, develop effective systems and strategies for effective communication of agricultural technologies
- To undertake development programmes for evolving more effective and efficient approaches, methods, and models in higher agricultural education for use in SAUs and Deemed Universities of ICAR
- To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- To develop expertise in information technology to meet the application software needs of NARS
- To undertake systematic review and to develop appropriate systems for efficient and effective administrative and financial management in NARS
- To function as a repository of ideas and information in the form of databases in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas
- To offer consultancy services in the field of agricultural research and education management, and agricultural information and communication technology.

4.3 Organization and Management

The Academy is headed by the Director. The Director is assisted by two Joint Directors (currently one Joint Director is in position) and three Division Heads in the execution and implementation of various programmes. The Academy is organized into three functional divisions, viz. Agricultural Research Systems and Policies Management, Human Resource Development, and Information and Communication Management. The Institute Management Committee (IMC) guides and supports the Director by periodic review of programmes and approval of investments in new areas of research and education, training programmes, workshops and seminars. The Research Advisory Committee (RAC) and Staff Research Council (SRC) provide broad guidelines and assist in developing and implementing specific research programmes and projects.

NAARM ORGANOGRAM



4.4 Human and Financial Resources

4.4.1 Human Resources (as on 31.03.2001)

Category	Sanctioned Strength	Posts Filled	Vacant Positions
1. Research Management Positions	03	02	01
2. Scientific	40	22	18
3. Technical	66	58	08
4. Administrative	48	47	01
5. Supporting	41	40	01
TOTAL	198	169	29

4.4.2 Financial Resources (as on 31.03.2001)

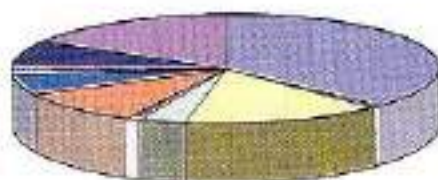
The budget allocation and expenditure during 2000-01 are given in the following table. Information presented in the table below provides details on finances available from other sources and revenue generated during the reported year.

Scheme	Head	Budget allocation (Rs. in lakhs)	Expenditure (Rs. in lakhs)
Non-plan	Establishment Charges	241.40	238.07
	Travelling Allowances	2.70	2.67
	Other Charges	97.80	94.35
	Works	-	-
	Sub-Total	342.00	335.09
Plan	Establishment Charges	25.00	23.85
	Travelling Allowances	6.00	5.91
	Other Charges	62.50	53.77
	Works	39.50	39.45
	Sub-Total	133.00	122.78
Other Sources	A.P.Cess	12.38	9.34
	AHRD	54.29	13.44
	NATP	104.58	9.36
	Sub-Total	171.25	32.14
	Grand Total	646.25	490.01

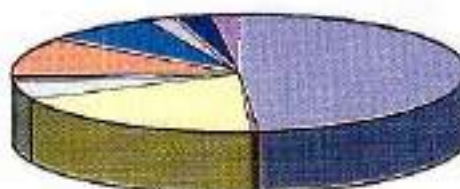
4.4.3 Income Generation

S.No	Item	Amount (in Rs.)
1	Training	12,57,098.00
2	Consultancy	2,250.00
3	Sale of Books, Technology, etc.	49,005.00
5	a) Room rent / License Fee	5,53,401.00
	b) Interest on Loans & Advances	2,61,736.00
	c) Leave salary / Pension consultancy	1,59,218.00
	d) Miscellaneous Receipts	5,79,144.00
	Sub-total of a to d	15,53,499.00
Sub-total of 1 - 5		29,84,372.00
6	Interest on Short Term Deposits	25,35,559.00
7	Sale of Vehicles / Machinery	1,11,800.00
Sub-total of 6 - 7		26,47,359.00
Total receipts		56,71,731.00

NAARM Budget (2000-01) - Allocation



NAARM Budget (2000-01) - Expenditure



■ Establishment Charges
 ■ Other Charges
 ■ Travelling Allowances
 ■ Works
 ■ AHRD

■ Travelling Allowances
 ■ Establishment Charges
 ■ Other Charges
 ■ A.P. Costs
 ■ NATP

5. Training Achievements

Imparting training in the broad area of agricultural research and education management is the key activity performed by NAARM. During the reported period, the Academy organized a total of 64 training programmes and trained 1,198 personnel belonging to scientific, technical and administrative cadres of National Agricultural Research System in India and some other developing countries.

5.1 NATP Sponsored Training Programmes

- **Priority Setting:** Relevant resource material for equipping agricultural scientists with necessary knowledge and skills on formal methods of prioritizing research programmes and projects has been developed. Utilizing the resource material developed, NAARM organized programmes on priority setting and trained 47 scientists of NARS.
- **Retreat Programme for the Senior Executives:** Through an institutional consultancy arrangement with IIM, Ahmedabad, a 'Retreat Programme for the Senior Executives (Top 20) of ICAR' was developed. It is planned to organize this programme in two phases, the first one at IIM, Ahmedabad and the second one at NAARM, Hyderabad. The consultancy proposal was submitted to NATP for approval.
- **Winning Research Proposal Writing:** A programme on 'Winning Research Proposal Writing' was developed through an institutional consultancy with ISNAR and submitted to NATP for approval of the competent authority.
- **Faculty Development:** Detailed programmes for training as well as for study visit by the faculty members of the Academy in premier management training institutes were developed and got approved by NATP/ICAR/World Bank. The faculty members were encouraged to initiate advance action to contact the host institutions abroad for their acceptance and fix up their programme schedule as per the guidelines provided by NATP.

5.2 AHRD Sponsored Training Programmes

AHRD provided support for building infrastructure, procuring equipment, developing educational technology for improving competency of teachers of SAU system. The training or infrastructure development is directed towards improving the quality of agricultural education.

Under this project extensive training to TNAU faculty on computer aided instructional technology was imparted. About 74 senior professors were trained for sensitizing on the development of Computer Aided Instruction. Th Deans, Directors and Principals were given training on leadership and management, while 55 administrative functionaries were trained on administrative management and computer applications.

A brief summary of the important programmes organized during the year is presented below:

5.3 International Programmes

Programme on Agricultural Research Management for Zonal Research Coordinators from Tanzania

This programme was organized from April 24 to June 2, 2000 with the support of the World Bank under TARP II for the nine zonal research co-ordinators of the United Republic of Tanzania. The major objectives of the programme were to expose the participants to various facets of agricultural research management and to analyse and synthesize Indian experience on specific areas of management by the participants for immediate application in their back-home situation. The programme was organized in two phases. In the first phase, which was conducted at the Academy, the participants were familiarized with four major capsules, viz. research project management, human resource management, information management, transfer of technology and communication, using interactive sessions in a participatory mode, by experts from the Academy, ICAR, and private agencies. In the second phase, the participants were taken, on a study visit, to various Agricultural R&D Institutions located within and outside Hyderabad to gather first-hand information on how agricultural research is being managed at the Institute level in India. During their study visit to New Delhi, the participants were provided with a unique opportunity to interact with senior executives involved in the policy planning at the ICAR Headquarters and the officials of ASRB responsible for recruitment and career development of scientists in the Indian NARS. At the end, the participants developed specific action plan, based on the analysis and synthesis of their learnings from the programme, for immediate application in their back-home situation. The programme was co-ordinated by Drs S.N. Saha, T. Balaguru and S.K. Nanda.



Programme on Agricultural Research Management for Directors from the Republic of Yemen

At the request of the Government of Yemen and the Food and Agriculture Organization, two months International Programme for two Yemenese Directors was organized from November 1 to December 30, 2000. During the first month of the training programme the trainee participants attended the English Language Course at the Central Institute of English and Foreign Languages, Hyderabad and Russels Institute, Hyderabad. The purpose of this course was to provide extensive coaching to the participants to gain proficiency in written and spoken English. In the second month i.e. from December 1 to 30, 2000 they attended the training programme in agricultural research management at the Academy. During the course, the participants were exposed to various facets of agricultural research project management, scientific communication, administrative and financial management. Computer awareness programme was exclusively designed to expose them to the fundamentals on the use of computers and to give them an insight into the working of the most commonly used programmes i.e. Microsoft Word, Microsoft Excel, Microsoft Power Point, E-mail, and internet. Drs J. C. Katyal and T. Balaguru coordinated the programme.



Training Programmes for Sri Lankan Officers and Scientists

Under the aegis of the Work Plan for 2000 - 2001 between the Indian Council of Agricultural Research (ICAR) and the Sri Lankan Council for Agricultural Research and Policy (CARP), the following training programmes were organized.

Programme on Administration and Financial Management

A training programme for Administrators and Accountants from Sri Lanka was organized from January 22 to 27, 2001. The major emphasis of the programme was on the facilitative role of administration and accounts towards improving the efficiency and effectiveness of research organizations through positive orientation, unbiased action and awareness of the value of time. Keeping in view the needs of the participants and the contents originally agreed upon, a variety of topics relating to the themes administration, financial and personnel management were covered. The participants evinced keen interest in some special lectures on purchase procedures in the ICAR system, project based budgeting, agricultural research finance information system and accounting and auditing procedures. The programme also served as a platform to share a common concern in fostering convergence of efforts of scientists, administrators and finance functionaries. The programme was coordinated by Drs T. Balaguru, M. Narayana Reddy and R. Kalpana Sastry.



Programmes on Agricultural Research Management

Two training programmes on agricultural research management were organized for the scientists from Sri Lanka. The major objectives of the programmes were to expose the participants to various principles of agricultural research management and to analyze and synthesize Indian experience on specific areas of management by the participants for immediate application in back-home situation. These programmes were organized in three main capsules viz. Agricultural Research Management, Human Resource Management, and Information Management. The first capsule covered topics such as research management process, technology forecasting, PRA techniques, research priority setting, project planning and formulation, project based budgeting, management of finance, project management techniques, farming systems approach, research extension linkages, privatization of agricultural research, monitoring and evaluation, intellectual property rights and world trade agreement. Second capsule covered interpersonal relationships, motivation of scientists, leadership development in R&D, stress management, time management and scientist-administrator interface. Third capsule covered internet and e-mail, use of information technology in agricultural research, database management concept, and application of geographic information system to natural resources management. The participants were taken on a study visit to various agricultural R&D institutions located in Hyderabad.

A total of thirteen Sri Lankan Scientists were trained in these programmes. The first programme was coordinated by Drs S.N. Saha, S. Shanmugam and S.K. Nanda, while the second was coordinated by Drs D. Rama Rao, Janardan Jee and B. S. Chandel.



5.4 National Programmes

Foundation Course for Agricultural Research Service (FOCARS)	
FOCARS 70 02.06.2000 to 29.09.2000	The FOCARS is the basic induction training imparted to all the newly recruited scientists to the agricultural research service. This four-month duration course is conducted in three specific phases namely, Phases I: Agricultural Research Systems and Management Processes - Concepts & Principles; Phase II: Field Experience Training and Phase III: Agricultural Research Systems and Management Processes - Practice and Applications. Phases I & III are organized at the Academy, and Phase II is conducted at the FET centres. The foundation course aim at exposing the participants to the various aspects of agricultural research project management, in addition to developing skills in them for managing their research by inculcating inter-disciplinary team spirit. Adequate opportunities were provided to the scientists to develop knowledge and skills in information technology, communication, organizational behaviour and administration and finance. This training also aims at developing a feeling of fraternity among the newly recruited scientists. NAARM faculty teams of Drs A. Gopalam & N. H. Rao; S. Shanmugam, S. K. Soam & B. S. Sontakki and P. Manikandan & B. S. Chandel coordinated the foundation courses organized during the year. A total of 182 newly recruited scientists were trained through these three courses.
FOCARS 71 25.08.2000 to 22.12.2000	
FOCARS 72 01.12.2000 to 30.03.2001	

Senior Level Courses

Statistical Software for Data Analysis 04.04.2000 to 15.04.2000 and 20.03.2001 to 31.03.2001	These courses aimed at familiarizing the participants to the advanced statistical software packages and to develop needed competency and skill in them to use these packages for quick, precise and effective data analysis, interpretation and reporting. Drs K. Vidyasagar Rao and M. N. Reddy coordinated these courses, which were attended by a total of 20 participants.
Management Development Programmes (3rd & 4th MDPs) 24.04.2000 to 29.04.2000 and 30.11.2000 to 06.12.2000	Management development programmes were held in response to the felt need for organizing courses in research management for the newly recruited heads of the divisions and project coordinators of ICAR institutes and headquarters. Keeping in view the management responsibilities of the incumbents in these positions, the programmes had the objective to provide a comprehensive overview of the various dimensions of research, human resources, information and financial management. A total of 31 participants were trained in these programmes, which were coordinated by Drs Jagannadham Challa and S. N. Saha.
Management Programme for Women Scientists 01.05.2000 to 06.05.2000	This programme was organized exclusively for the women scientists of the NARS. It addressed a variety of issues confronting the women scientists in becoming equal contributors to agricultural R&D. It was coordinated by Drs Janardan Jee and N. Sandhya Shenoy and attended by 17 women scientists of ICAR/SAUs.
Improving Administrative Management in ICAR 30.05.2000 to 09.06.2000	This course was organized to enable the participants to develop an appreciation and understanding of various issues of administrative management in the ICAR system, with a view to foster higher work ethics, positive attitude to work and team spirit to achieve organizational goals. Twenty nine participants attended the programme, coordinated by Dr S. N. Saha and Mr. B. Pandu Reddy.

Training Achievements

Computer Applications in Agricultural Research 20.06.2000 to 01.07.2000; 12.12.2000 to 23.12.2000 and 13.02.2001 to 23.02.2001	Three programmes were organized during the year. The subject matter of these courses aimed to provide exposure and adequate hands-on experiences to the participants on the basics and applications of computers in agricultural research. The team of Drs. K. Vidyasagar Rao and K. M. Reddy coordinated these programmes. A total of 33 participants including four from NAARM were benefited by these courses.
Rapid, Relaxed and Participatory Appraisal and Participatory Learning and Action Techniques for Research and Extension in Agriculture 20.06.2000 to 01.07.2000	The objective of this programme was to impart the requisite knowledge and skill in gainfully using the various participatory methodologies for research and extension in agriculture. This programme was attended by 17 participants and coordinated by Drs. V.E. Sabarathnam and N. Sandhya Shenoy.
Executive Development Programme in Agricultural Research Management 15.07.2000 to 18.07.2000 and 02-02-2001 to 05-02-2001	Two executive development programmes were organized to provide a forum for the Directors of the ICAR institutes to discuss various research management issues confronting the effective functioning of the ICAR institutes. It gave an opportunity to the participating Directors to share their experiences based on individual case and specific situations. In addition, it exposed the participants to a few practical tools for solving research management problems. These programmes were attended by 18 Directors of ICAR institutes and coordinated by Drs. S. N. Saha and Jagannadham Challa.
Agricultural Scientist Development for Personal and Organizational Effectiveness 17.07.2000 to 28.07.2000	The objectives of this programme were to examine different organizational structures, cultures and processes, and to get acquainted with organizational information systems and technologies and to adapt the model and approaches of the behavioural sciences in their back-home situations. The programme dealt with HRD strategies for agricultural research, personal effectiveness attributes, interpersonal relationships and transactional analysis, management of work stress, performance appraisal, management of creativity and problem solving, group behaviour and group dynamics project, management of differences, information systems, effective communication, vigilance and disciplinary procedures, and team management. Eleven scientists / faculty from ICAR Institutes and from SAUs participated in the programme. Drs. S. Shanmugam and Jagannadham Challa coordinated the programme.
Educational Video Production 08.07.2000 to 29.07.2000 and 12.12.2000 to 23.12.2000	Two programmes were organized on this topic during the year aimed at equipping the participants with necessary knowledge and skills in planning and production of educational video programmes. These courses are fully hands-on, and by the end of the programmes the participants produced video programmes for use in educational programmes. A total of seven participants were benefited by these courses, which were coordinated by the team of Drs. R. C. Chelvan, Janardan Jee and Mr. K. R. Prabhakar.

Training Achievements

Instructional Aids for Effective Presentations 3.07.2000 to 9.07.2000	This course was organized with the objective to impart knowledge and skills in planning, production and use of instructional aids for effective presentations. Two participants attended the course coordinated by Drs N. Sandhya Shenoy and Janardan Jee.
Analytical Tools and Techniques for Assessing Agricultural Sustainability 01.08.2000 to 06.08.2000	This training programme attended by seven participants was organized to introduce the various analytical tools and methods to define, quantify and assess sustainability of agricultural production systems and to provide hands-on training in the use of above tools and methods. Drs N. Hanumantha Rao and M.N.Reddy coordinated the programme.
Agricultural Research Prioritization Techniques 21.08.2000 to 26.08.2000 and 18-01-2001 to 24-01-2001	These courses were organized with the objectives to familiarize the scientists with the various techniques used for prioritizing agricultural research projects and to share with them the practical experiences of priority setting at different levels. It dealt with the policy context and need for research prioritization, participatory rural appraisal methods to understand the problems faced by the farmers, and the techniques of short listing and prioritizing the research problems. Forty six scientists from ICAR institutes and SAUs, who are connected with the work of agricultural research prioritization and those connected with the preparation of strategic research extension plans, participated in the course. Drs K.P.C. Rao and B. S. Chandel coordinated these courses.
Participatory Methods in Agricultural R&D 11.09.2000 to 16.09.2000	This course conducted in workshop mode was designed to deliberate on the use of various participatory methods for agricultural research and development. Checklists for conducting various participatory methods were developed in the interactive sessions, besides providing hands-on experiences in using the participatory methods. The course, coordinated by Drs N. Sandhya Shenoy and VE. Sabarathnam, was attended by seven participants.
Learning to Lead and Succeed in New Environment 14.09.2000 to 20.09.2000	This course was designed to provide necessary skills on personality and leadership development to the middle level scientists of ICAR and SAU systems, so that they can effectively take up the responsibilities of research and educational management. It was attended by four participants and coordinated by Drs Janardan Jee and Jagannadham Challa.
Computer Awareness Programme 18.09.2000 to 23.09.2000	This programme was offered with the objective of providing basic knowledge and skills in effective use of computers in agricultural research for the beginners. It was coordinated by Dr. K. M. Reddy and attended by 19 participants.
Video Editing Techniques 19.09.2000 to 29.09.2000	This course aimed at providing hands-on training in the post production editing techniques for the video programmes. It also covered the basics of planning and production of video programmes. Six participants participated in this course coordinated by Mr K. R. Prabhakar.
Impact Assessment of Agricultural R&D 25.09.2000 to 30.09.2000	This programme was conducted with the objective of developing an understanding and appreciation of the impact assessment of the agricultural R&D, as follow up measure. It dealt with the various methods and tools of impact assessment. It was coordinated by Drs B. S. Chandel and K. P. C. Rao and attended by 15 participants.

Training Achievements

Project Management Using Microsoft Project 30.10.2000 to 04.11.2000	This course aimed at developing the knowledge and skills of participants in using Microsoft Project for efficient and effective research project management. It covered the concept and tools of research project management with emphasis on MS-project tools in effective monitoring of the project resources. Fifteen participants participated in this course, which was coordinated by Dr. D. Rama Rao.
GIS Applications in Agriculture 06.11.2000 to 16.11.2000	Realizing the growing applications of GIS in agricultural research, management and extension, this training programme was organized to orient the participants to GIS tools and their applications in agriculture. This was an introductory programme. Thirteen scientists/ faculty of ICAR/ SAUs with basic knowledge of computers attended the programme, which was coordinated by Drs M.N. Reddy and N. Hanumanth Rao.
Agricultural Research Project Management 14.11.2000 to 24.11.2000	This course was organized to equip the participants with knowledge and skills in various aspects of agricultural research project management. It was coordinated by Drs S. K. Nanda and D. Rama Rao, and attended by 11 participants.
Indian Seed Industry Management 20.11.2000 to 25.11.2000	This course targeted exclusively to the R & D personnel involved in Indian seed industry aimed at developing understanding and appreciation on various management issues of Indian seed industry. It was attended by 22 participants and coordinated by Drs K. P. C. Rao and Janardan Jee.
Advanced Data Analytical Techniques for Research and Management 28.11.2000 to 08.12.2000	This programme was organized to familiarize and create confidence among the trainees to analyze the data using various computer interactive analytical softwares such as statistical, graphical and simulation modeling and provide state-of-art analytical tools to 'optimize the information'. Eight scientists/ faculty of ICAR/SAUs with basic knowledge of computers participated in the programme, which was coordinated by Drs M.N. Reddy and K. Vidyasagar Rao.
Designing and Developing Web Pages 17.01.2001 to 19.01.2001	The first web designing course in NARS was organized with an objective to study relevant web sites for information and use in web designing. The programme dealt with creating web pages, applying themes, formatting text, creating internal and external links, creating navigation bar, inserting images, hyper linking images, creating tables, enhancing tables, adding web extras using the Front Page. Selected extension and education web sites and Agricultural Gateway of India (AGI)/ Agricultural Technology Information Centre (ATIC) prototypes were demonstrated. Twenty scientists from the ATIC centres participated in the course coordinated by Drs N. Sandhya Shenoy and M. Narayana Reddy.
Techniques of Effective Linkages in Agriculture 14.03.2001 to 21.03.2001	This programme was organized for the benefit of scientists, and extension personnel to orient them to develop understanding of inter-system linkages and equip them with techniques of measuring and strengthening linkages. The topics covered in the programme related to prospects and problems of linkages in NARS, linkage issues, partners in agricultural development, indicators of effective linkages, identification of linkages and measurement, matrix mapping, case analysis of inter-system linkages, stake holder analysis, terms of reference, strategies and techniques of linkage development. Seven scientists of ICAR/ SAUs attended the programme, which was coordinated by Drs VE Sabarathnam, V.K. Jayaraghavendra Rao and B. S. Sontakki.

Management Programme for Women Scientists 19.03.2001 to 24.03.2001	This programme was organized to develop knowledge and skills of women scientists for managing their scientific activities and to provide required managerial skills to work effectively as leaders and members of different teams. It was organized in four main capsules, viz. agricultural research management, gender issues in research organization, women and resource development and behavioural dynamics. Twenty-four middle and senior level women scientists from ICAR/SAUs participated in the programme, which was coordinated by Drs Janardan Jee and N. Sandhya Shenoy.
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Sponsored Programmes

Faculty Development Programmes (FDPs) in Educational Psychology and Behavioural Technology for the TNAU. 10.04.2000 to 20.04.2000; 29.05.2000 to 03.06.2000 and 11.09.2000 to 21.09.2000	A series of FDPs in various areas are being organized by the Academy under the aegis of the AHRD and as a part of the MOU with the Tamil Nadu Agricultural University, Coimbatore. During the year three such programmes were organized in the area of educational psychology and behavioural technology for the benefit of middle and senior level faculty members of TNAU. These programmes were organized in an effort to reorient the agricultural education to meet the changing demands and needs. They specifically aimed at equipping the participants with the requisite knowledge and skills in applying the various concepts and principles of educational psychology and behavioural technology to maximize teaching learning effectiveness in agricultural education. A total of 56 faculty members of TNAU were trained in these programmes, which were coordinated by NAARM faculty teams of Drs A. Gopalani, Jagannadham Challa, R. Kalpana Sastry, P. Manikandan, P. Kumar and T. Balaguru.
FDP in Behavioural Technology 26.08.2000 to 06.07.2000	This programme was also conducted as a part of the MoU between NAARM and TNAU. Its objective was to orient the teachers of TNAU to the concepts and principles of behavioural technology in order to effect behavioural modifications to optimize teaching learning effectivity. This programme was coordinated by Drs A. Gopalani, P. Manikandan and S. Shanmugam and attended by 22 teachers of TNAU.
Orientation and Reorientation Programmes on Educational Administration and Management of Financial Issues 29.04.2000 to 04.05.2000; 17.07.2000 to 27.07.2000 and 07.11.2000 to 17.11.2000	These programmes were specially and exclusively designed for the functionaries of administration and finance wings of TNAU to achieve total quality. Various dimensions related to agricultural education, economic reforms, personnel and financial management, personality development and other general administrative and financial issues formed the curriculum of these programmes. NAARM faculty Drs K. P. C. Rao, A. Gopalani, D. Rama Rao, Janardan Jee, Capt. Mehar Singh and Mr. B. Pandu Reddy coordinated these programmes, in which a total of 80 personnel belonging to the administrative and finance wings of TNAU were trained.

Training Achievements

Prioritizing Research Programmes 18.01.2001 to 24.01.2001	The main objective of this programme was to infuse in them adoption of tools and techniques assisting in arrangement of various research projects in order of their priority. A total of 27 scientists belonging to diverse subject areas and representing Indian NARS were exposed to formal methods of prioritizing research programmes and projects. Initial feedback confirmed acceptance of the subject focus and course content. While it will take some time to assess adoption of priority setting tools and techniques, these programmes did sensitize the participants' mindset on the need to prioritize where resources are scarce, research projects are many and the outcomes have to be right and relevant. Dr T. Balaguru coordinated the programme.
Winter School on Decision Support Systems in Agriculture 25.09.2000 to 15.10.2000	Effectiveness and impact of decision-making depend on the timely and quality decisions taken. Decision support system plays an important role in effective and efficient planning and management of agricultural research and education. With the tall future challenges to be achieved in agricultural research and educational development, there is a need to sensitize the faculty members, researchers, and scientists of the NARS to the importance and use of decision support systems. In realization of this need, the academy, with the financial support of ICAR organized a winter school from September 25 to October 15, 2000. Twenty two scientists and faculty members from various ICAR institutes and SAUs participated in the programme. Drs D. Rama Rao and S.K. Nanda coordinated the programme.

5.5 Feedback

One of the striking features of the programmes organized by the Academy is feedback on various aspects of the programmes, gathered through a structured questionnaire. Comments / suggestions, thus received, are collated and synthesized into recommendations for revising prevalent methodologies/approaches on organization and delivery of various courses to introduce future improvements. The following table summarizes the feedback received on various programmes, conducted by NAARM during the period under report.

Name of the programme	Feedback received
Statistical Software for Data Analysis	- Programme was good and the practical sessions were organised in a befitting manner - New advanced packages are not included
MDP in Agricultural Research	- Examples of survey data should be included along with SPSS - More Panel discussions should be included in future - Theme-application of information technology in resource management is extremely useful - Vigilance procedures may include disciplinary action aspects and more time for vigilance theme - Time allocation for various theme areas was good - Needs improvement and update the materials
Management Programme for Women Scientists	- More time is desirable for the topics like finance, purchase procedures, etc. - Good exposure to different areas - Quality resource material was supplied - More computer exposure required - Some exercises in behavioural aspects to be given - More women guest faculty required - More such programmes should be organised
Improving Administrative and Financial Management in ICAR	- Topics were dealt by the experienced faculty - Good opportunity for learning new skills
Computer Applications in Agricultural Research	- Content and the training material were good - Still more time needed for practical sessions - Softwares were relevant and useful - Individual computers to each participant needed - Duration should be three to five days more
Rapid Rural and Relaxed Participatory Rural Appraisal and Participatory Learning Techniques for Research and Extension in Agriculture	- Learnt many new techniques in PRA - Training should be extended by four more days, and more time shall be allocated for village work - More number of guest speakers to be invited
Gahan Hindi Prashikhan va Karyashala	- Refresher courses should be organised - More practical training than theory - Training on computers for working in Hindi
EDP in Agricultural Research	- Case study exercises to be included - More time for disciplinary and administrative matters and also financial rules and procedures - Need video show on relevant topics - Should be one month - Brainstorming sessions on management topics relevant to ICAR should be included

Training Achievements

Agricultural Scientist Development for Personal and Organization Effectiveness	- Group seminar presentations were good - Content is good and relevant
Educational Video Production	Still more hands-on experience needed
Instructional Aids for Effective Presentations	- Organization of the programme was good - More time needed for video editing and photography - Duration may be increased more
Analytical Tools and Techniques for Assessing Agricultural Sustainability	- Good learning experience - More practicals on GIS and other techniques shall be covered in the course
Participatory Methods in Agricultural Research and Development	- Still more time for village practicals - Duration may be increased to 2-3 weeks - Partnership with NGOs to utilize their expertise for village visits
Computer Awareness	- Exercises were very well organized and training material supplied was interesting and useful. - Duration should be extended by one week - More assistance needed during practicals - Planning and practical utility for administrative staff
Video Editing Techniques	- Duration should be one month - Practical editing experience was extremely good
Impact Assessment of Agricultural Research and Development	More practical exercises to be included
Decision Support Systems in Agriculture	- Topics on neural networks and interfacing GIS with expert system can be added - More training material needed - Educational visits may be arranged
Project Management Using Microsoft Project	- Link between time and resource management in project management, emphasis on practicals, presentations and teaching aids are strong points - Costing and budgeting of resources in project management can be improved - More guest lectures could have been arranged
GIS Applications in Agriculture	- A topic on establishing the GIS laboratory need to be established - Extend the GIS facilities to ICAR institutes and SAUs - Relevant to the needs - Course content and coverage is good
Agricultural Research Project Management	- Case studies needed - More time for practicals - Well structured and content was good
Advanced Data Analytical Techniques for Agricultural R&D	Windows based analytical tools were well practised

AHRD Sponsored Programmes	• Duration of these programmes should be increased • Need more time for practical • Usefulness of group discussion • Useful training material provided • Commitment and expertise of the faculty members • Relevance and need of the programme • Tighter schedule of the programme • Good educational experience • Impressive method of teaching
International Programmes	• Acquisition of new knowledge and skills • Good educational experience • Course content, modus operandi, and practical utility of the programmes very satisfactory • Need for more study visits to institutions out side Hyderabad • Refresher courses may be planned in the future to other colleagues • More time needed for computer practicals • Need to provide additional training material like publications, etc.

6. Research Achievements

The Academy scientists conduct research in the areas of agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. Funding for some of the projects is through AP Cess, DFID and NATP. In addition, the Academy also supports short-term projects aimed at developing specific case studies for use in its programmes. A list of the projects carried out during 2000-01 is detailed below:

S. No.	Project Title	Funded by	Investigators
1	Management parameters of state agricultural universities and emerging policy issues and implications	AP Cess	A. Gopalani
2	Developing instructional modules for gender in agriculture curriculum	AP Cess	A. Gopalani
3	HRD strategies for organizational effectiveness in NARS	AP Cess	Jagannadham Challa K. Vidyasagar Rao
4	Micro-level research priority setting: Strategic research and extension plan (in collaboration with MANAGE)	NATP	K. P. C. Rao, B. S. Chandel, N. Sandhya Shenoy, S. K. Soam, V. K. Jayaraghavendra Rao, B. S. Sontakki
5	Performance assessment and accountability enhancement	NATP	T. Balaguru
6	Development of computer mediated courseware for facilitating learning - teaching	NATP	A. Gopalani
7	Distance training for agricultural research management. (In collaboration with ISNAR, COL & Wye College)	DFID	Nicole Brunet, ISNAR (up to March 2000), Paul Perrault, ISNAR (from April 2000 - March 2001) and Jagannadham Challa
8	Assessing impact of oilseed research in India	NAARM	B.S. Chandel
9	Role of government organization and non government organization run krishi vigyan kendras in agricultural development	NAARM	N. Sandhya Shenoy K. Vidyasagar Rao
10	Organizational climate in ICAR institutes	NAARM	M. M. Anwer
11	Studies on application of ITK in animal management for enhancing production	NAARM	P.Kumar, K.V.S.Rao N. Sandhya Shenoy
12	Implementation of farming systems research in India: Case studies	NAARM	T. Balaguru P. Manikandan
13	The reaction of farmers of Andhra Pradesh and farm graduates of ANGRAU towards the proposed agri-clinics and agri-business centres	NAARM	J. C. Katyal, V.E. Sabarathnam, N. Sandhya Shenoy, B. S. Sontakki, V. K. J. Rao, S. K. Soam

Major activities and accomplishments of the projects during the year 2000-01 are briefly given below.

6.1 AP Cess fund projects

Project-1 Management parameters of State Agricultural Universities and emerging policy issues and implications

Objectives

- To study the ICAR model act and its implications in various agricultural universities.
- To probe into the acts, statutes and functioning of management bodies of SAUs.
- To ascertain the organizational structure of universities and their physical development.
- To investigate the integrated function of teaching, research and extension within the universities, and allied institutions within the state and the state departments of agriculture.
- To outline the course credit and internal assessment systems in various agricultural universities.
- To review investment pattern in SAUs and the outcomes.
- To scrutinize the recruitment and staffing pattern vis-à-vis the load of work and the outcome.
- To understand the planning, monitoring and evaluation mechanisms in the SAUs.
- To specify recommendations as a result of the above eight objectives and enlist the policy resulting out of this study.

Progress

A comprehensive document both hard copy and CD was developed by developing a structured questionnaire, and was field validated for content criteria and purpose by piloting it to some selected respondents. An interview schedule was preceded for the research data in some selected universities.

Project-2 Developing instructional modules for gender in agriculture curriculum

Background

A research scheme with funding from A.P. Cess on developing instructional modules for integrating gender in agricultural curriculum was initiated during February 2001. The back ground of initiating such a research scheme is that, urban graduates are reluctant to serve in rural areas, more number of rural students be admitted in the agricultural courses so as to increase the number of professional graduates with rural back ground, who may take up extension related jobs / activities. Women reservations are extended to all activities supported by state such as scholarships, admissions, and nominations to awards, curricular and co-curricular activities in order to ensure the gender equity in all developmental activities relating to agriculture. It is required that the organizational climate in the line departments needs to be made gender-friendly to encourage women to take up extension jobs and the women employees are to be provided necessary facilities and services like transport, accommodation and spouse employment, etc., in order to ensure more women participate in the over all social development.

Research Achievements

It is proposed to use the instructional resource modules as course-ware in Under Graduate instruction and thereby sensitize on the gender role in agriculture after initial validation and testing. The developed instructional modules will have the components for which adequate research work is contemplated.

Progress

Module development strategy was developed in order to bring in uniformity in the developed modules. Integration of text with multimedia effects is being worked out.

Project-3 HRD strategies for organizational effectiveness in NARS

Objectives

- To develop management systems approach for enhancement of organizational effectiveness.
- To develop and standardize case models for selected institutions for research productivity and personnel development of individuals.

Progress

- A series of eight brain storming sessions were held with the middle and senior level scientists across the Indian NARS to supplement the data being collected from five selected ICAR institutes.
- Using the data collected so far from different institutions of NARS, specific databases were developed on personality profile, management training needs, HRD climate, etc.

6.2 National Agricultural Technology Project (NATP)

Under the aegis of Organization and Management (O&M) Reforms component of National Agricultural Technology Project (NATP), two sub-projects, namely: (a) Institutionalization of Priority Setting, Monitoring and Evaluation and Networking of Social Scientists (to be implemented jointly by NAARM, NCAP and IASRI), (b) Human Resource Development for Agricultural Research and Education Management were sanctioned to NAARM for implementation during 2000-2003.

Major activities and accomplishments of the NATP projects during the year 2000-01 are briefly presented:

Project-4 Micro-level research priority setting: Strategic Research and Extension Plan (SREP)

Objectives

- To develop appropriate research priority setting methods at the micro level
- To train the scientists in micro-level priority setting

Progress

- By utilizing the resource material developed by NAARM, two training programmes of one-week duration were organized during August 2000 and January 2001, which were attended by 47 scientists from ICAR Institutes and SAUs.
- In collaboration with MANAGE, the SREP methodology was refined and SREPs were prepared for seven districts in Andhra Pradesh, Bihar, Himachal Pradesh, Maharashtra and Punjab.

Project-5 Micro-level research priority setting: Performance assessment and accountability enhancement**Objectives**

- To strengthen NAARM's capacity to deal with the issues of instructional performance and accountability through collaborating research, case studies and training
- To introduce and institutionalize improved performance measurement system in a small number of research institutes
- To derive generic lessons from this work for incorporation in NAARM's training and advisory programmes.

Progress

- Preparation of base papers on the tools and techniques for performance assessment and accountability was initiated
- Action was taken to enter into institutional consultancy with ISNAR, The Netherlands, to work on this component
- During the recent visit to ISNAR, the basic framework for the tools and techniques to be used in the study were finalized.

Project-6 Development of computer mediated courseware for effective learning - teaching**Objectives**

- To develop interactive learning modules in various agricultural subjects
- To develop educational technology learning modules for use in distance education

Progress

- Computer assisted instructional material for three modules in educational technology terms and concepts, elements in soil sciences and management of people were developed.
- Field testing of second module is in progress at TNAU, Coimbatore.

6.3 ICAR / NAARM - ISNAR Collaborative Project

Project-7 Distance training for agricultural research management

This ISNAR-ICAR/NAARM research project, funded by Department for International Development, Government of United Kingdom and ISNAR, The Netherlands, is carried out in collaboration with the Wye College, UK and the Commonwealth of Learning, Canada. The project aims at developing and standardizing a distance training methodology on agricultural research management to enhance the contribution of NARS to sustainable development leading to alleviation of rural poverty.

Progress

- > Based on the data collected in the first phase, the applied research, of the project an applied research report of the project and a communication directory of NARS were published and released.
- > In the second phase of the project, action research, a workshop was conducted from 9 to 19 January 2001 at NAARM to finalize the structure and content of distance training course. The workshop was attended by Mr. Derek Ray, expert from Wye College, UK, Dr Paul Perrault, Project Manager from ISNAR and Dr Jagannadham Challa, Principle Investigator of the project from NAARM. Based on the discussions held during the workshop, a draft course material consisting of five modules was prepared.
- > The complete training package including self learning modules, support reading material, worksheets, etc. was developed.
- > For implementing the distance course in the action research phase, the necessary logistics, delivery and support systems were finalized.

6.4 Institute projects / Case studies

Project-8 Assessing impact of oilseed research in India

Objectives

- > To estimate change in productivity and cost structure of oilseed crops
- > To determine contribution of research and policy support to oilseeds productivity
- > To evaluate return to investment of oilseed research in major oilseed crops in India.

Progress

- > The data collection on area, production and productivity was completed.
- > Information on cost of production and input use in different crops across various states gathered.
- > The analysis work on first two objectives is completed. The total factor productivity was worked out for two crops – groundnut and rapeseed and mustard.
- > The write-up and drawing of inferences is in progress.

Project-9 Role of government and non government organization run krishi vigyan kendras in agricultural development**Objectives**

- To study, compare and assess the selected organizational, workgroup, and job involvement variables and roles performed in agricultural development by the GO and NGO KVKs
- To note the expectation of the select clientele groups of the GO and NGO KVKs in view of changing agricultural scenario
- To find the inter-institutional linkage patterns among the NGO KVKs and other line departments, their expectations, operational constraints and suggestions for meaningful collaboration

Progress

- The sample covered a total of six KVKs – three each of NGO and GO KVKs established up to 1995, selected randomly from three regions of Andhra Pradesh state viz., Coastal, Telangana and Rayalaseema.
- Three types of interview schedules were developed separately for data collection from the KVK in-charges, KVK staff and the clientele groups.
- Under each KVK, a total of 30 respondents from the client groups representing 10 farmers, farm women and farm youth were selected through stratified random sampling methods.
- The data collection from all the existing staff of the selected KVKs and the clientele groups was completed.
- The data compilation and analysis is under progress.

Project-10 Organizational climate in ICAR institutes**Objectives**

- To generate base line data on organizational climate in ICAR institutes.
- To suggest measures for improving the organizational climate.

Progress

- A pilot study was completed in selected institutes located in Hyderabad.
- The analysis and the subsequent refinement of the methodology is in progress.

Project-11 Studies on application of ITK in animal management for enhancing production**Objectives**

- To compile information on the ITK related to animal management
- To document and translate them into major Indian languages for use by the developmental agencies and farmers.

Research Achievements

Progress

- The basic information was compiled through the Field Experience Training programmes undertaken by the scientist-trainees of foundation courses.
- The information so gathered was compiled and grouped into various themes.
- The broad themes were digestive disorders, reproductive disorders, infectious diseases, parasitic diseases, other diseases, wounds and injuries, animal management practices, and proverbs and beliefs.
- The information was presented state-wise under each theme.
- This compilation was published in the form of a technical bulletin entitled "Native Traditional Wisdom for Livestock Management".

Project-12 Implementation of farming systems research in India: Case studies

Objectives

- To investigate the FSR methodology followed by one research institution and to suggest suitable alternatives for effective implementation.
- To develop case studies that serve as resource material for training in FSR.

Progress

- Agricultural research institutions in the NARS are currently focussing their attention to Farming Systems approach for technology generation. In order to understand this aspect in a greater detail, a case study was developed based on the information collected from the Agricultural Research Station at Aruppukottai of TNAU. A number of 'Integrated Farming Systems' involving crops, livestock and horticultural crops have been developed and assessed for their feasibility, productivity, profitability and sustainability. Among the integrated farming systems developed, the one involving crops (Cotton + Blackgram), livestock (Tellicherry goat) and horticultural crops (pomegranate and bet) was found to be the most suited to the rainfed areas. Issues associated with the implementation of the programme by the research station scientists were also analysed.
- Based on the above experience a case study was developed for use in the training programmes organized by the Academy.

6.5 Adhoc Survey

Project-13 Reaction of farmers of Andhra Pradesh and farm graduates of ANGRAU towards the proposed agri-clinics and agri-business centres

Background

- This ad-hoc survey was carried out at the behest of the Council, to know the opinions and reactions of the actual beneficiaries - farmers and farm graduates - of the farm clinics and agri-business centres proposed to be started by the Government of India in the Tenth Plan.

- This proposal aims to provide self-employment to the farm graduates and thereby supplement the extension and service efforts of the State Departments of Agriculture.
- Two separate surveys were planned and conducted during October 2000 to elicit opinions and reactions of farmers and farm graduates.

The salient findings are as follows

- All respondent farmers opined that farm clinics and agri-business centres should be started in villages by professionally qualified graduates specialized in agriculture, horticulture, veterinary, home science, etc.
- A sizeable percentage of the farmers desired that agricultural graduates should start the farm clinics. No definite trend could be observed regarding the level of expertise required to start the farm clinic.
- The farmers were ready to pay for the farm clinic services up to Rs 50/- per visit, though a sizeable number of respondents said that they would like to pay Rs 10/- for each farm clinic visit.
- With regard to the nature of service expected by the farmers from the farm clinic, various types of services emerged from the study which were advisory, input supply, diagnostic, identification and control of pests and diseases, therapeutic, marketing and supply of relevant literature. The top most two services desired by farmer respondents were advisory and input supply, and the least being literature supply.
- The farmers opined that provision of transport facilities, adequate equipment, intermittent training, latest information, quality inputs, regular time based contacts and rendering correct solutions for farmers problems would pave way for better extension system which would also be applicable for the proposed farm clinic and agri business centres.
- A good amount of heterogeneity was observed regarding the effect of the proposed farm clinic and agri business on the current extension system and input supply agencies. The respondents perceived monopoly of agri input sale and information service. Competition and complementarity were the possible effects that the proposed farm clinics would have on the current extension system and input supply agencies.
- The farm graduates opined that they would go for consultancy when they would have no other option but prefer a government job.
- Majority of the farm graduates (44%) wanted to establish farm clinics as mono clinics as an individual venture. Twenty seven per cent of the farm graduates were interested to have agri clinics as poly clinics, i.e. as a multi-specialty group activity.
- About 40 per cent farm graduates were confident to establish farm clinics and agri business centres. The farm graduates opined that they needed training (52%) in order to make them fully competent. Surprisingly, more than 40 per cent of the farm graduates could not judge themselves whether they needed competency building or not.

Research Achievements

- The farm graduates needed training in specific technical areas for further competence especially 'hands-on field experience'. The farm graduates expressed training needs in olericulture / floriculture, dairy, garment manufacturing, consumer education, farm produce marketing, etc.
- Majority of the farm graduates reported that they would like to have livestock health service centres and dairy enterprise. Plant nursery, poultry export oriented business, tissue culture unit, vermiculture, vegetable, post harvesting, agri business, compost making, garment making, mushroom sawn production, consultancy services, bio-agents production and pest control services including veterinary and pathological laboratories were various types of enterprises preferred by the farm graduates in establishing the farm-clinics and agri-business centres.
- Financial requirement was Rs. five lakhs, according to majority of the farm graduates to start the farm clinic. Most of the graduate respondents wanted credit requirement for establishing farm-clinics and agri-business centres.
- Nearly half the farm graduates felt that this is a new venture, and they should be given subsidy up to 50 per cent. The farm graduates wanted that there must be three to five years of moratorium period in loan repayment as they needed some time to establish their business.
- Surprisingly, sizable percentage of the farm graduates (44%) wanted to establish the agri enterprise at district headquarters and only 30 per cent of the farm graduates wanted to establish the enterprise in their own village.

7. Achievements in Policy Support

7.1 ICAR Training Policy

The Academy took initiative to make financial provision for Human Resource Development (HRD) activities, and it was reflected in the ninth plan budgets of all ICAR institutes. Subsequently, academy developed training policy for ICAR, which provides guidelines for institutionalization of HRD activities in the institutes. Training policy strives to shift focus from the present problems to the planned future, provides a platform to clarify organizations procedures and approaches, acts as a guidance and support for the leadership, and finally, acts in disseminating abstract vision statements into implementation thereby linking vision to action. Thus, a positive and vision-based planning of human resources is expected to help the organization regroup and create collaborative environment in which functional teams (administration, technical and scientific) focus on desired outcomes.

Forecasting HRD needs is an important strategic tool for organization to ensure that its progress is aligned with organization missions. The proposed training policy recommends creation of a cell for coordination of HRD activities to facilitate the achievement of institutional, departmental and individual goals through training needs analysis. The cell in turn undertakes an in-house training need analysis to evaluate the context, content, users, suitability to resolve identified performance problems and a cost-benefit analysis. Some broad features of the training policy is described below.

Training logistics: If the training is to be effective it has to be organized at a location away from work place. This condition can be relaxed in case of women employees but they too need to be encouraged to go for outstation training. HRD team needs to look into technical aspects as well as physical arrangements part.

Budget: HRD team can identify and approve proposals to engage specific skill-based consultants to train its staff in a highly specialized area. The present mind set that HRD budget is only to meet travel and registration fee cost need to be changed. Diversity in implementation of HRD budget is essential to meet location-specific HRD needs. HRD team needs to be empowered to utilize the HRD budget.

Induction training: In principle, every employee on the first entry to council must undergo induction training. This need is being fulfilled for administration and scientific personnel at officer level. Council needs to evolve policy and arrangement to train technical and administrative personnel at lower entry levels also. This is vital for the organization to be effective. As induction training aims to orient the new entrants to the system, its mandate, vision and procedures, it is essential to be carried out within ICAR in the lines of scientists training.

Refresher training: To meet the UGC/ICAR policy on career advancement of scientific staff, all major institutes and select departments in universities need to be encouraged to organize more courses with support from education division of ICAR.

All non-scientific cadre officers also need to be trained once in five / seven years. This training is essential to improve skills in specific functional areas, inculcating team spirit and vision sharing, behavioural skills and to orient to changes in organizational procedures and policies. Besides, such refresher courses improve communication across system and individuals.

Skill development of ministerial and technical staff: Large mass of personnel working in ministerial and technical positions need to be sent for skill based trainings. The HRD team can identify the skill development areas for their institute and also identify institutions / agencies that provide such training. Each such training should include use of IT in day-to-day work, and this is to be given importance in all training programmes.

Web-based training: Web-based interactive multimedia is an effective and flexible way to provide training. This tool permits quick updation of the content for wider reach at no extra cost. The biggest hurdle in web-based training is high start-up costs for investment in technology and development time. ICAR need to encourage development of web-based training and also motivate employees to use it.

HRD Audit: It is possible to provide bench marks to make audit easier and bring transparency in the HRD effort. Assessment would include extent of maturity of HRD system, strategies, competencies, style, cultures and linkages. ICAR to follow this mechanism to put in right track its HRD efforts.

7.2 A Manual on Purchase Procedures in ICAR System

Background: In order to streamline and develop simplified Purchase Procedures, ICAR constituted a Committee under the chairmanship of DDG (CS). This committee inter alia recommended that NAARM should develop a Manual on Purchase Procedures to be followed by the ICAR System. In pursuance of that decision, NAARM analyzed the provisions contained in the GFR vis a vis all possible literature on the existing purchase procedures followed in ICAR or other similar autonomous research organizations.

General purchase procedures: Presently, for procurement of recurring and non-recurring stores, ICAR and its constituent institutes/projects follow rules and guidelines given in the GFR. Inflexible nature of GFR topped with mandatory adherence to its provisions, frequently leads to delays putting hotfoot to scientific experiments, which by nature are time and season-fixed. Keeping in view the time-sensitivity, specialized requirements and the unique status/importance accorded to scientific contribution, the Government of India (GOI) on recommendations of the Scientific Advisory Committee had delegated enhanced powers and exempted scientific organizations from certain provisions of GFR. In response to the relaxations given by the GOI, several scientific organizations have developed simpler guidelines on effecting diverse purchases, but within the provisions of GFR. From time to time, ICAR has also made attempts to streamline the purchase procedures to be followed by its constituent institutes.

Past Initiatives on purchase procedures: In 1983, ICAR initiated the process of reforms in purchase and procurement of stores by enhancing the delegated powers of institutes Directors. Enhancement of powers enabled Directors to approve purchases at the institute level. While this reform accelerated the process of purchases, it by no means was

a step in the direction of simplification of purchase and procurement procedures. It was only in 1988 when ICAR set up a committee under the chairmanship of Dr A.M. Michael to evolve and justify the introduction of alternative purchase procedure in harmony with the GFR. The committee made several recommendations, which at best can be described as advisory in nature, since they emphasized more on laying out the specifications before stores are purchased and on inspection after the procurements are made. Consequently, in practice the ICAR and its institutes continued to adhere strictly to the guidelines enunciated in chapter eight of GFR.

In 1995, Dr O. S. Tomar (the then Director NDRI, Karnal) and his associates proposed simplified purchase procedures, which were presented as a supplementary agenda during the meet of the Directors of ICAR institutes. Classification of purchases into five separate categories (i.e., stationery, chemicals/consumables, equipments, emergent/shutdown purchases and special items like non-standard items and those whose prices fluctuate widely) and constitution of corresponding purchase committees to focus on individual procurement of items so classified was the main thrust of the measures suggested by Tomar et al. Introduction of 'Concurrent Auditing' in place of prevalent 'Pre-Auditing' step was another highlight of the recommendations. For want of formal follow-up, the simplified purchase procedures proposed by Tomar and his group remained un-adopted.

It was in 1997, when Mr G. Parthasarthy, the then Financial Advisor addressed a letter to all the Directors of ICAR institutes on the subject 'Planning for Procurement of Stores'. Along with this letter, Mr Parthasarthy appended general instructions on store purchase steps. His treatment of the subject was, thus, far the most direct and simplified narration of the GFR related to purchases. However, his treatment of the subject was so brief that it required further elaboration to detail out various purchase procedures against the backdrop of provisions contained in the GFR and subsequent instructions issued by GOI/ICAR.

Rationale for a manual on purchase procedures: Despite several past efforts, concerns on adopting detailed, simplified but foolproof and transparent purchase procedures continue to be an integral aspect of ongoing organization and management reforms. The interest in the matter is very genuine, since significant portion of the Council's budget is utilized for the procurement of consumable and non-consumable stores. And purchases made with diluted application of terms laid out in GFR, attract adverse observations by the statutory audit. Additionally, such purchases are open to allegations on misuse of authority, mismanagement of public funds, and corruption.

The purchase procedures enunciated in GFR are adequate in many ways. By and large, they are difficult to corrupt if followed in letter and spirit. Though they are not always fast enough to operate. Whether true or false, less sensitivity to speed calls for an unbiased introspection to understand if the delay related concerns are rooted in GFR or they stem from certain external factors. Balance of evidence contained in various reports (Michael committee report, Tomar's committee recommendations or G. Parthasarthy's instructions on purchase of stores) or Manual on Purchase Procedures developed by CSIR univocally suggest that delays are primarily a result of improper planning, which include:

1. Non-projection of items in the Institute EFC (absence of administrative approval).
2. Shift to unapproved items.
3. Indents without proper justification and/or specifications of stores intended to be purchased.

4. Change of specifications before/after the orders are placed.
5. Purchases in the face of existing inventory.
6. Insistence on purchases of stores of proprietary nature without adequate back-up information/certification.
7. Delay in receipt of funds.

Proper planning also includes allowing enough lead-time to complete pre- or post-purchase processes. Purchases made in a hurry often lead to non-adherence to, or dilution of GFR regulations. Compromising on certain provisions of GFR to accommodate haste in formalizing the purchases is an often-committed error. Buying of stores of value exceeding rupees two hundred thousand by limited quotations in place of open tender is a typical example of rush purchases made by ignoring the conditions of GFR. Changing the agreed upon (between vendee and vendor) tender conditions/specifications of goods to be supplied to accept whatever is readily available is another example of effecting swift purchases, albeit in contravention of rules and procedures. Whether it is breach of set guidelines or it is mutation in terms and conditions of contract following purchase procedures are completed/orders are made is a serious violation of GFR. Hence, there is an urgent need to develop a purchase manual giving guidelines on appropriate and suggest, wherever necessary, simplified purchase procedures within the ambit of GFR. Development of a purchase manual is also necessary so that various ICAR Institutes follow uniform purchase procedures with common interpretation of GFR.

The Purchase Procedures Manual drafted by NAARM is focused to maximize procedural-transparency and price-competitiveness and to minimize the operational difficulties while effecting purchases of diverse items of various kinds. In fulfillment of this objective, it stresses to transact by decentralization of purchases through appropriate purchase committees with built-in mechanism of concurrent auditing, while retaining the independence of pre-auditing step. So that purchases not only conform to technical specifications as per indenter's requirements, but are also in tune with the latest development in instrumentation and electronics, the concept of vetting by a technical committee is suggested. In order to move towards on-line purchase operations suggestions are incorporated to make them computer-friendly. Introduction of varied color tabular indent/purchase forms to accommodate indents/purchase orders/pacts of different kind is the innovation suggested to replace the prevalent system of note sheets. Additionally, it includes copies of all the relevant procedures and guidelines issued by the Government of India/ICAR from time to time on the subject of purchase procedures.

In general, the NAARM Manual on Purchase Procedures fulfills the following objectives:

- To outline the GFR guided purchase procedures, which infuse transparency in application,
- To enhance competitiveness in prices without lowering the quality of the goods being purchased and are easy to operate.
- To define various categories of stores and the related purchase procedures
- To outline the constitution and role of purchase/technical committees.
- To suggest computer-friendly indent/purchase procedures.

The final draft of the manual entitled "Purchase Procedures in ICAR System" has been submitted to ICAR for approval.

7.3 Constitution, Functions and Powers of Institute Management Committee

In 1976, Indian Council of Agricultural Research (ICAR) approved creation of an Institute Management Committee (IMC) for each of the research institute under its purview. The purpose was to grant autonomy in the matters of administration and management of the affairs and environs of the Institute so as to support approved research plans. It was a modest beginning in the direction of decentralization. Subsequently, in 1995 when the powers of the Directors of the ICAR Institutes were enhanced, functioning of IMCs was also strengthened. Following that, some existing powers were enhanced and a few more were granted. Despite these genuine efforts in support of institutional autonomy, dependence on ICAR headquarters for the clearance of several crucial administrative and financial matters remains as strong as ever. In pursuance of this persisting concern, the subject on role and powers of IMCs was discussed during the 1998 meetings of the ICAR Directors. It was decided that NAARM should review afresh the entire issue of functions and functioning of IMCs. Consequently, in 1999, through a structured questionnaire, NAARM sought opinion of Directors of ICAR Institutes on the current and envisioned working of IMCs. Out of 93 Directors contacted, 44 responded.

Based upon (1) the synthesis made from the input given by 44 Directors in response to the above mentioned questionnaire, (2) review of the existing powers sanctioned by the ICAR to IMCs, and (3) a review of functions and powers granted to Management Committees/ Councils/ Boards by research organizations enjoying similar status as ICAR, following suggestions are made on improving the constitution, functions and functioning (by grant of additional administrative/financial authority) of IMCs.

Constitution of IMC

Although current constitution of IMC is found adequate in intent and content, the existing set up, however, needs strengthening from functional and contributive angles. A critical re-look into some aspects of IMC constitution, following changes are suggested (Table 1):

Functioning of IMC

Periodicity of meetings: Holding of two meetings a year of the IMC should be made mandatory; organization of more number of meetings than two should be need based and decided by the Chairperson.

Quorum: As it prevails, if a minimum of four members is present during a meeting, officially the quorum is deemed to be complete. It is suggested that participation of at least two (out of four) members from outside the host Institute should be made compulsory to satisfy the official requirements on quorum.

Table 1. Proposed changes in current constitution of IMC

Rule	Membership/status	Proposed change
66(a)(1)	Director of the Institute - Chairman	Status quo
66(a)(2)	A representative of the State Government in which Institute is located, nominated by the President of ICAR - Member	Official need to be from a mid-cadre
66(a)(3)	A representative of the other State Government concerned with the research of the Institute, nominated by the President ICAR - Member	Vice-Chancellor/Director/ Dean of an SAU may be a preferred choice
(66)(a)(4)	A representative of the Agricultural University having jurisdiction over the area, nominated by the President ICAR - Member	Vice-Chancellor/Director/ Dean of an SAU
(66)(a)(5)	Two non-official persons representing agricultural/rural interest, nominated by the President ICAR - Members	May belong to a local NGO, genre of progressive farmers or private sector.
(66)(a)(6)	Four scientists of the Council's Institutes, nominated by Director General ICAR - Members	Number may be raised to 6; 4 drawn from host institute, 1 each from another ICAR institute and different scientific organization (CSIR etc.)
66(a)(7)	A Council's representative, nominated by Director General ICAR - Member	Status quo
66(a)(8)	The Financial Advisor of the Council/DARE or the Accounts Officer of the same or another institute - Member	Status quo
66(a)(9)	Administrative Officer of the Institute - Member secretary	Status quo

Role and functions of IMC

In order to maximize contribution of IMC in institutional affairs, a clear definition to role to be played and delineation of functions to be performed will be essential. ICAR has already enunciated the functions of IMC. Since every function of IMC is preceded by the word phrase 'consideration of,' means 'a body of appointed persons delegated to consider, take action on or report on' some matter, it becomes necessary to reformulate and reiterate the

various functions of IMC. In the first place, it will be necessary to rename it as 'Council', which will allow it to act as an administrative body and exercise managerial control on affairs of the institute. Hence, in order to widen scope and role of current 'Institute Management Committee', it is proposed to rename it as 'Institute Management Council'. In the second place, renaming itself will not be adequate until an alternative schedule of functions is accepted and appropriate powers are delegated to perform those functions. In fulfillment of that objective, it is suggested to strengthen the existing list of functions by appropriate modifications and by addition of some more functions. While doing that, it will be necessary that the new list of functions, harmonizes broadly the accomplishment of institutional work plan with efficiency and effectiveness and serves the cause of functionaries who execute it.

Based on a critical analysis of the: (1) existing functions assigned to IMCs of ICAR Institutes, (2) functions and powers of the Executive Council of MANAGE, and (3) functions of the Management Council of CSIR Laboratories, alternative functions proposed to be performed by the future IMCs are given in Table 2.

Table 2. Existing and proposed functions of IMC

Existing functions	Proposed functions
Consideration of proposals for Five Year Plan and Annual Plan.	To regulate and manage the attainment of goals set for each research program approved by Research Advisory Committee and to monitor and evaluate their progress.
Periodical review of progress of development schemes.	To guide while preparing five-year and annual budgetary outlays and approve purchase/hire/take on lease immovable (land/buildings) or movable (equipments or machinery) property.
Consideration of proposal for annual budget.	To guide and support institutional affairs aimed at improving the working environment in the institute.
Consideration of items of expenditure, which are beyond the powers of the Director.	To approve proposals on write off/disposal within the purview of extant rules.
Policy issues relating to the Institute, including the rights and obligations of the staff.	To approve execution of contracts for security and consultancy (by institute scientists and out-sourcing for institute building) and enter into research collaboration agreements.
Consideration of action taken on the recommendations of the Grievance Cell and Institute Joint Council.	To grant approvals on some miscellaneous expenditure items, which need observance of strict economy measures.

Financial and administrative powers

The IMC needs to be extended financial and administrative powers that enable it to accomplish the various functions given to it by the ICAR. The prevalent financial and administrative powers of IMCs are moderate in extent and routine in nature. By and large, the current powers merely allow IMC to reaffirm application of that what has

already approval of ICAR. What little authority has been granted hardly gives necessary freedom to IMC to take independent decisions that fully support institutional cause. Typically, the clause-mandating chairman to submit, along with proceedings, a certificate to ICAR that 'no decision of IMC is in violation of sanctioned powers' binds him to play safe. Consequently, functions of IMC though decentralized in letter are centralized in spirit.

Apparently, there is a need for an in-depth re-look into the functions and role of IMC. The primary function of IMC is to govern and manage the affairs and working environment of an Institute. It should facilitate accomplishment of that goal in such a way that its resolutions support RAC-approved research/education (including HRD activities)/extension programmes of the Institute. Additionally, decisions made by the IMC ought to balance organizational/Institutional and employees' interest (on service matters/benefits/conflicts) with minimum procedural hassles. In order to create an enabling environment for self-sustaining decisions, IMC requires powers that raise its status and importance of role. Dispersion and delegation of more powers to IMC by ICAR, from now onwards, becomes necessary. A detailed document outlining the schedule of existing and proposed delegation of administrative and financial powers has been submitted for approval of the ICAR.

7.4 Gender Policies

A policy paper is prepared for ICAR highlighting policy initiatives needed for creating a conducive environment for the professional growth and satisfaction of agricultural women professionals in Indian NARS. The report is based on outcome of Academy's DST sponsored project on Issues of Professional Women in Agriculture.

The number of employed agricultural women graduates is less than five per cent of total agricultural graduates. The number of female students entering agricultural education is on the rapid rise in the recent past, i.e., proportion of female students increased from about four per cent in 1980 to about 25 per cent in 2000. Further, in some southern states male and female students are equal in proportion. Considering large influx of female graduates in to employment in the coming years, there is an urgent need to reorient the systems to accept women graduates in the workforce.

Some important issues that affect the career of professional women in agriculture relate to their mobility, spousal employment, flexibility in work hours, physical facilities and gender discrimination in recruitment, selection, etc. Considering the factors affecting the agricultural women graduates in India, policy initiatives needed for creating a conducive environment for their professional growth and satisfaction are suggested.

The mobility of scientists between university and ICAR institutes is low at less than 0.5 per cent. It is desirable that at least five per cent of the total scientific staff move to other systems so as to infuse new ideas and new work relations to sustain the research pace. Thus, there is a need to increase the mobility through other

means. ICAR being an apex body to provide policy guidance to the two systems, it can bring policy changes to improve the mobility of scientists between the two systems.

For more efficient and effective utilization of the work force, the study points to certain policy initiatives. Major problem of working women professionals in agriculture relate to the placement and mobility - especially when both respondent and spouse are working. Much of the potential of professional women in agriculture could not be harnessed due to spousal displacement. Increasing mobility among different systems would greatly reduce this problem. Since ICAR and SAUs work together, there is need to increase mobility of scientists between the two systems. ICAR, being the apex body and single largest employer in the field of agriculture, should take lead in initiating policies aimed at improved mobility of scientists between the two systems and influence the State Agricultural Universities (SAUs) to follow the same.

The following policy initiatives are recommended for action

- There should be a policy for free exchange of scientific personnel in the same cadre between the SAUs and ICAR. Deputation may be allowed for the scientific manpower from the SAUs and ICAR in the same cadre for a period of five years. This would improve the mobility between the systems and stem the spousal displacement. During the leave period, the institutes may be permitted to hire persons on contract basis giving preferential treatment to spousal employment.
- Option to work reduced hours can be given to young mothers. This will enable them to cope with office and mother-hood responsibilities and also reduce demands for long leave to take care of young children.
- Institutes be encouraged to evolve and implement employee friendly policies and human resource development must be structured into plan activity. Institutes can provide transport to work place and run creche with trained staff and make it available to employees as a welfare activity.
- Preferential allotment of accommodation to the single women employees would help in providing security and reduce problems arising out of family displacement.
- At least one female member may be associated in all official committees such as recruitment, selection, transfer, etc. This would not only minimise gender-based discrimination but also enhance their participation, and it would give them an opportunity to express their needs and difficulties.

8. Consultancy Achievements

Over the years, the Academy has gained expertise in various areas of agricultural research and education management. The faculty members of the Academy take up consultancy projects and services, depending upon the specific needs received from the client organizations. Consultancy services of varied nature were offered by the faculty members over the last few years. This also includes management support provided to ICAR as and when needed. The consultancy services extended during the year under report are briefed below.

- The Academy has completed the consultancy projects of video production for Central Plantation Crops Research Institute (CPCRI), Kasargod. The video productions are 'CPCRI at a Glance' and 'Integrated Management of Pests and Diseases of Coconut'. The completed video tapes have been handed over to CPCRI and have been well appreciated by them. At present the Academy has taken up another two similar projects for Directorate of Rice Research, Hyderabad and Project Directorate of Biological Control, Bangalore.
- Dr J.C. Katyal, Director, NAARM worked as a Senior Research Fellow at Centre for Development Research, University of Bonn, Germany, from June 5, 1999 to June 1, 2000. During the period he prepared a concept paper on desertification and made a critical review on organic matter status of tropical soils. Dr Katyal also worked as a teaching faculty for the international doctoral programme organized by Centre for Development Research (ZFF), Germany.
- Dr T. Balaguru, Principal Scientist, undertook a consultancy assignment for two weeks (October 8 to 22, 2000) to assist the M.S. Swaminathan Research Foundation in preparing a developing plan for the Union Territory of Pondicherry on the biovillage model of sustainable agriculture and rural development. The main focus was on arresting natural resource degradation through the integrated management of soil, water, flora and fauna, as well as poverty alleviation by promoting multiple livelihood opportunities among the resource-poor in rural areas for additional income and employment generation.
- Dr Jagannadham Challa, Principal Scientist, undertook the draft manuscript preparation of the collaborative project work in the field of agricultural education under the aegis of AHRD project of Tamil Nadu Agricultural University for quality improvement of education. The studies were : a) Role of Heads of Divisions in Improving the Quality of Education in TNAU (consultant Dr S.R. Subramanian) and b) Studies on Quality Improvement Needs of Agricultural Education of TNAU (consultant Dr A.V. Rangarajan)
- Dr D. Rama Rao provided consultancy to DFID India in the area of assessment of higher education in Andhra Pradesh (AP). The study focused on impact and expenditure review of higher education in AP, so as to develop policy on public expenditure to match with the thrust areas on higher education reflected in AP Vision 2020.
- Dr K. P.C. Rao provided consultancy to MANAGE, Hyderabad for technical advice and assistance relating to "Orientation of state level officers on methodology for identification of demand driven research agenda in Assam".

9. Honours and Awards

- > **Rajabhasha Puraskar:** The Academy won the Rajashri Tanden Rajabhasha Puraskar of Indian Council of Agricultural Research (ICAR), New Delhi, for the year 1999-2000 for the best work done in the implementation of the official language in the 'C' region (Ga Kshetra). The award was presented by Shri Jagdambi Prasad Yadav, Hon'ble Member of Parliament at a special function held on October 16, 2000 at Krishi Bhavan, New Delhi. Dr J.C. Katyal, Director, NAARM received the award. The members working in the hindi section viz. Dr A. Gopalam, Officer-in-charge (Hindi); Mr D. Venkateswarlu, Technical Officer (Hindi); Ms J. Renuka, Assistant Director (O.L) and Mr Pradeep Singh, Assistant Director (O.L) were awarded commendation certificates on this occasion.



- > **Hyderabad Rose Society Challenge Cup:** The Academy won the prestigious Hyderabad Rose Society Challenge Cup for the highest number of aggregate points in various sections of roses, at the XIX All India Rose Show of Indian Rose Federation and Silver Jubilee Rose Show of Hyderabad Rose Society on January 7, 2001. The Academy also won Deccan Chronicle Challenge Silver Cup for the highest aggregate points in different variety of roses; T. Chandra Sekhar Reddy Memorial Challenge Silver Trophy for the best of yellow roses, besides winning six first prizes, ten second prizes and six third prizes. The Academy was awarded the second prize under the category of Central Government Institutional Gardens, in the eleventh Horticultural Show organized by the Department of Horticulture from January 26 to 28, 2001.
- > **ICAR Inter-zonal Sports Tournament:** The NAARM sports contingent, which participated in the final Inter-zonal tournaments held at Central Institute of Fisheries Education, Mumbai, from February 21-23, 2001, secured seven prizes. In the athletic events (women), as in the previous tournaments, Ms Rukmini Ammal bagged the first prizes (third consecutive year) both in discus throw and shot put. Ms G. Aneeja bagged first prize in javelin throw, second prize in shot put and third prize in long jump. In the athletic events (men), Mr M.K. Samson bagged the third prize in discus throw.
- > **Publication Award:** A research article authored by Drs J. C. Katyal, M. N. Reddy and K. L. Sharma published in Fertilizer News received the FAI Second Best Paper award for the year 1999.



10. Workshops, Conferences, Seminars Organized

10.1 Seminar on Protected Agriculture

A seminar on Protected Agriculture was organized at the Academy on May 10-11, 2000, to provide an overview of protected agriculture in horticulture crops and to share international experiences on commercial vegetable production in glass/poly-houses under protected environment. Dr Merle H. Jensen, Arizona University, Tucson, USA, attended the seminar as a resource person and shared his experiences on protected agriculture. He presented a detailed analysis of glass and poly house technologies in vogue around the world. He also explained the growth and protection of plants in such controlled environments and the type of low cost poly houses that can be built indigenously in India. Sixty delegates from SAUs, ICAR, state departments and private sector participated in the seminar. Dr D. Rama Rao coordinated the seminar.

10.2 National Conference on System Dynamics for Agriculture and Development

A Ninth National Conference on System Dynamics for Agriculture and Development was organized by the System Dynamics Society of India (SDSI) at the Academy during December 26-30, 2000. Dr Subba Rao, Vice Chancellor of Acharya N.G.Ranga Agricultural University, Hyderabad, was the Chief Guest of the conference and was chaired by Dr J.C. Katyal, Director, NAARM. Systems thinking is the need of the hour for agriculture and development, said Dr I.V. Subba Rao, while inaugurating the conference. Dr Katyal spoke on the need for greater interaction between different disciplines for developing



natural sciences such as agricultural sector. He was of the view that the situation of surplus management could have been successfully solved had there been systems thinking and approach. Professor P.K. J. Mohapatra, President of SDSI, gave a historical view of the evolution of systems thinking and system dynamics. The origin has been from natural scientists to represent and learn the natural complexities and behaviour of systems with time, in the long run. The delegates (40) included scientists and officials from ICAR institutes, Indian Institutes of Technology, Tata Consultancy Services, National Institute for Management of Agricultural Extension, Indian Institute of Management, Agricultural Universities, and NAARM. Dr D. Rama Rao coordinated the conference. A two-day pre-conference training on system dynamics preceded the conference.

10.3 Workshop on Official Language Policy Resolution of New Millennium

A two day National Workshop on Official Language Policy Resolution of New Millennium was organized on November 28 and 29, 2000 for the Hindi officers, Hindi translators, Hindi assistants and scientists-in-charge of Hindi sections of ICAR institutes.

A blend of lectures on the use and practical difficulties in the implementation of Official Language Policy of Government of India and lively discussions among the participants on various difficulties and future plans were among the high lights of the workshop. Based on their experience and as a consequence of discussion, the participants gave various practical suggestions for better use of official language in the new millennium.

The topics covered during the workshop were the importance of language and its changed form, the importance of language in the implementation of official language, cultural advancement with the use of language, constitutional provisions for the use of official language, New Millennium's expectations in the implementation of official language, national resolution for implementation of official language, etc. Forty seven members attended the workshop, coordinated by Dr A. Gopalam and Mr D. Venkateswarlu.

10.4 Workshop on Gahan Hindi Prashikshan

Gahan Hindi Prashikshan va Karyashala was organized from January 22 to 27, 2001 for the employees of ICAR headquarters and ICAR Institutes, who are working in the sections other than Hindi. The major objective of the workshop was to discuss about the experiential usage of Hindi. More stress was given on practical exercises in order to enable the participants to learn the use of Hindi in discharging their official duties in Hindi.



The workshop exposed the participants to the official language implementation; responsibilities and expectations of central government employees, different forms of correspondence, the difficulties faced by non Hindi speaking employees in the implementation of official language policy, and some tips to increase use of Hindi in the central government offices etc. Forty three members attended the workshop. Dr A. Gopalam and Ms J. Renuka coordinated the workshop.

10.5 Brainstorming on Good Governance in Agricultural Research Organizations

One-day Brainstorming session was organized at the Academy by the southern chapter of the National Academy of Agricultural Sciences (NAAS) on December 2, 2000 with the objective of developing an understanding of good governance and examine, how it can be worked into the organization and management of agricultural research in India. Dr J.C. Katyal, Director, NAARM, initiated the proceedings of the session by welcoming the invitees and making a formal presentation of the base paper entitled Good Governance in Agricultural Research Organization - ICAR Case. He identified the new challenges facing agricultural research systems and emphasized the need for good governance of research by drawing attention to the dwindling output from research in India with reference to science in general and agricultural science in particular. Starting with the definition of governance, he outlined the characteristics of good governance, and clarified how it extends beyond management, he suggested that core values

Workshops, Conferences, Seminars Organized

espoused by UNDP for good governance are also applicable to ICAR. The following issues were initiated by the fellows of NAAS. Autonomy in function of ICAR; decentralization and devolution of powers for autonomy of institutes service and finance rules - how to merge them with the value system; accountability - how it can be made part of work culture; team building and teamwork and human resources development. Nineteen fellows of National Academy of Agricultural Sciences and faculty members of NAARM attended the session. Drs J.C. Katyal, and N. Hanumantha Rao coordinated the programme.

10.6 Deans Workshop on Development of Curriculum for Postgraduate Degree in Seed Science and Technology

A workshop on the development of curriculum for postgraduate course in seed science technology in which 38 Deans from various state agricultural universities have participated. In this workshop a comprehensive curriculum was developed considering the over-all view point of the participating deans who were of the opinion that post graduate curriculum should have an upward linkage with the under graduate curriculum. The curriculum developed will be adopted by SAUs after scrutiny by the academic councils.

10.7 NATP Workshop

Relevant resource material for equipping agricultural scientists with necessary knowledge and skills on formal methods of prioritizing research programmes and projects has been developed. Utilizing the resource material developed, NAARM organized a sensitization workshop from August 21 to 26, 2000.

The main objective of the workshop was to infuse in them adoption of tools and techniques assisting in arrangement of various research projects in order of their priority. In all, 20 scientists belonging to diverse subject areas and representing Indian NARS were exposed to formal methods of prioritizing research programmes and projects. While it will take some time to assess adoption of priority setting tools and techniques, these programmes did sensitize the participants' mindset on the need to prioritize where resources are scarce, research projects are many and the outcomes have to be right and relevant.

10.8 SRC Meeting

The Staff Research Council (SRC) Meeting was held on August 18 and 19, 2000 to review and strengthen the research and training activities of the Academy and also to identify the institutions for overseas training or study visit of the faculty members during 2000-01 under the HRD component of NATP.

11.1 Visits Abroad

- Dr M. Narayana Reddy, Principal Scientist and Dr R. Kalpana Sastry, Senior Scientist, visited Asian Institute of Technology (AIT), Bangkok, Thailand from May 3 to August 1, 2000, to participate in the FAO Fellowship Training on educational technology applications in agriculture under AHRD Programme. The training aimed primarily to update the knowledge and skills in relation to principles, concepts, and innovations in education and training technology. During their stay at Thailand, Dr Reddy and Dr Sastry visited various agricultural universities, study centres and extension departments to understand and assess the applications of educational technology in teaching and extension.
- Dr. S.N. Saha, Joint Director, visited the Plymouth University in the United Kingdom on study tour for a period of three weeks from November 20 to December 8, 2000 under the AHRD programme. During the visit, he worked on 'educational management with particular reference to distance education'.

11.2 Participation in Professional Programmes

- Dr Jagannadham Challa, Principal Scientist was nominated as a special nominee on the Institute Animal Ethics Committee of the National Institute of Nutrition, Hyderabad since June 2000 for the purpose of Control and Supervision of Experiments on Animal (CPCSEA), Ministry for Social Justice and Empowerment, Government of India, New Delhi.
- Drs V.K. Jayaraghavendra Rao and Bharat S. Sontakki, Senior Scientists, participated in the National Seminar on private extension at MANAGE, Hyderabad on July 28 and 29, 2000 and presented a paper entitled "Research-Extension Linkages - Case of a Private Seed Industry".
- Dr S.K. Soam, Senior Scientist, participated in the National Seminar on Private Extension at MANAGE, Hyderabad on July 28 and 29, 2000 and presented a paper entitled "System Analysis of Private Extension".
- Dr Jagannadham Challa, Principal Scientist, was deputed by ICAR along with Dr Mruthyunjaya and Mr. Md Aslam to participate in the deliberations held under the auspices of Study Group I of the Estimates Committee of Parliament on September 4, 2000 at Panchmarhi, Madhya Pradesh. The deliberations were held under the chairmanship of Hon'ble Minister of Parliament Dr U. Venkateswarlu.
- Drs V.K. Jayaraghavendra Rao and Bharat S. Sontakki, Senior Scientists, participated in the SREP Workshop held at Adilabad from September 18 to 25, 2000.

- ✱ Dr B.S.Chandel, Senior Scientist was invited to be associated with MANAGE in prioritization of researchable problems of Strategic Research and Extension Plan (SREP) under NATP. In this context, he visited District Kangra, Himachal Pradesh, from November 4 to 12, 2000, one of the 24 districts selected for SREP on pilot basis.
- ✱ Dr J.C. Katyal, Director, attended the Fertilizer Association of India (FAI) annual seminar entitled Agricultural Subsidies – Global Dimensions, held from November 11 to 13, 2000, at Hotel Ashok, New Delhi.
- ✱ Dr J.C. Katyal, Director, participated in the Vice Chancellors Conference on November 22, 2000, at Krishi Anusandhan Bhavan, New Delhi.
- ✱ Dr J.C. Katyal, Director, participated in the National Symposium on Accreditation for Quality Assurance in Agricultural Education on November 23, 2000, at Krishi Anusandhan Bhavan, New Delhi.
- ✱ Dr R. Kalpana Sastry, Senior Scientist, participated in International Seminar on "Protection of India's Intellectual Wealth in the New millennium" on December 2 and 3, 2000 at India International Center, New Delhi.
- ✱ Dr P. Manikandan, Principal Scientist, participated in the programme on Networking of Agricultural Information Centres, organized by MANAGE, Hyderabad on December 14 and 15, 2000.
- ✱ Dr J.C. Katyal, Director, attended the 88th Session of the Indian Science Congress held at IARI, New Delhi from January 3 to 7, 2001.
- ✱ Dr J.C. Katyal, Director, attended annual dialogue on Information and Communication Technology for Poverty Eradication at M.S. Swaminathan Research Foundation, Chennai from January 19 to 21, 2001.
- ✱ Dr B.S.Chandel, Senior Scientist was invited in January, 2001 to review ISNAR Research Management Guidelines on 'Strategies and Actions for Improving University Contribution to National Agricultural Research'.
- ✱ Dr P. Manikandan, Principal Scientist, participated in the Workshop on Networking of Libraries, organized by Acharya N.G. Ranga Agricultural University, Hyderabad on March 7, 2001.

11.3 Invited Lectures Delivered

- ✱ Dr D. Rama Rao, Principal Scientist, was invited as guest faculty on Internet and its Use at WALAMTARI, Hyderabad during October, 2000.
- ✱ Dr R. Kalpana Sastry, Senior Scientist, was invited as guest faculty on IPR and Indian Agriculture at Institute of Chartered Accountants of India, Hyderabad Chapter during July and October, 2000.

- * Dr R. Kalpana Sastry, Senior Scientist, was invited as guest faculty at Tea planters' Association Assam Branch at TOCKLAI research institute, Jorhat on October 5, 2000.
- * Dr J.C. Katyal, Director, delivered Dr N.S. Randhawa Memorial Lecture entitled Organic Matter Maintenance: Main Stay of Soil Quality at Punjab Agricultural University, Ludhiana, organized by the Indian Society of Soil Science, Ludhiana Chapter on October 31, 2000.
- * Dr J.C. Katyal, Director, delivered Dr Ch. Krishna Moorthy Memorial Lecture entitled Desertification – Myths and Realities on December 22, 2000 at Bhaskara auditorium, Birla Science Centre, Hyderabad.
- * Dr S.K. Soam, Senior Scientist, delivered a lecture on Prioritization of Technological Options for Integrated Livestock and Forage Production to the participants of "Workshop cum Seminar on Dryland Agriculture" on January 25, 2001.

11.4 Higher Degrees Awarded

- * Dr R. Kalpana Sastry, Senior Scientist, completed successfully a distance learning course on Intellectual Property Rights offered by World Intellectual Property Organization (WIPO), Geneva.
- * Mr K.R. Prabhakar, Programme Officer (CCTV), was awarded Postgraduate Diploma in Human Resources Management (PGDHRM) by the Indira Gandhi National Open University (IGNOU), New Delhi.

12. Publications

12.1 Research / Review / Conference papers

Chandel, B.S. 2000. Agricultural research - emerging perspective in the context of new economic policies. Paper published in National Seminar on "WTO and its Implication to Agriculture" organized by MANAGE at Shimla on 27 and 28 July, 2000.

Chattopadhyay, C. and Kalpana Sastry, R. 2000. Methods for screening against sesame stem-root rot disease. *Sesame and Safflower Newsletter* No. 15: 68-70.

Chowdhary, V.M., Rao, N.H., and Sarma, P.B.S. 2000. Use of geographic information systems in groundwater flow modelling. Proc. National Workshop on rainwater and groundwater management for rice ecosystem, Indian Institute of Technology Kharagpur, India, September 25 and 26, 2000.

Jagannadham Challa and Thulasamma, L. 2000. HRD strategies for developing corporate culture in veterinary and agricultural sciences. Paper presented in AGRI INTEX - 2000 held at TNAU, Coimbatore from August 29 to September 3, 2000.

Jagannadham Challa. 2001. Why the need for alternatives for animal experimentation? Paper presented on the occasion of the WHO sponsored Pre-clinical Toxicology Workshop, 2001 held at NIN, Hyderabad from February 5 to 10, 2001.

Jagannadham Challa. 2001. HRD Strategies for integrating agricultural sciences with corporate business management, paper presented in the National Workshop on February 25 to 27, 2001 held at Gulbarga University, Gulbarga.

Jayaraghavendra Rao, V. K. 2000. Research-extension linkages in private sector - case of a private seed industry. Paper presented in the national seminar on private extension at MANAGE, Hyderabad on July 28 and 29, 2000.

Jha, A; Patel, A.A., and Rao, K.H. 2000. Kinetics of Physico-chemical changes in instant kheer mix powder during storage, 14th convention of ICFOST on Modern trends and perspectives in food packaging for 21st century held in Mysore from November 22 to 24, 2000.

Joshi, P.K., Chandel, B.S., Virmani, S.M., and Katyal, J.C. 2001. Agricultural performance of semi-arid tropics. *Indian Journal of Agricultural Economics*, Vol. 56 (1): 71-88.

Kalpana Sastry, R. and Chattopadhyay, C. 2001. Effect of soil solarization on wilt incidence and seed yield of safflower. *Indian Phytopathology* (communicated).

Kalpna Sastry, R., Rama Rao, D., Kresur V.R., and Vijaya Kumari, R. 2000. Systems analysis of safflower production in India. 5th International safflower conference, July 22-24, 2001, Williston, North Dakota, USA.

Karthikeyan, J., Sushil Kumar, Anjeneyulu, A.S.R., and Rao, K.H. 2000. Application of hurdle technology for the development of caprine keema and its stability at ambient temperature, *Meat Science* 54 : 9-15

Katyal, J.C. and Narayana Reddy, M. 2001. Fertilizer situation in south asia. Contributions to the Encyclopedia of Life Support Systems (EOLSS), USA.

Katyal, J.C. and Rao, N.H. 2000. Good governance in agricultural research organization – Case ICAR. Paper presented at the Brainstorming Session on Good governance in agricultural research organizations, held at NAARM, organized by southern chapter of NAAS, on December 2, 2000.

Katyal, J.C. and Paul L.G. Vlek. 2000. Desertification – Concept, Causes and Amelioration, Centre for development research (ZEF), University of Bonn, Germany.

Nirmale, Vivek H., Sontakki, B. S., and Biradar, R. S. 2001. Indigenous technical know-how of coastal fisher-folk of Sindhudurg district in Maharashtra. Paper presented in the national symposium on fishery technologies and their commercialization, CIFE, Mumbai & IFA, Mumbai (January 11 and 12, 2001).

Rama Rao, D., Vijaya Kumari R. and Hanibabu, E. 2000. Agricultural education in India: A sociological perspective. *Outlook on Agriculture*, 29(3), 177-184.

Rama Rao, D., Vizayakumar, K., and Narsinghani, S. 2000. Design and analysis of demand and supply for edible oil in India. Rama Rao, D. (Ed). Application of system dynamics for development in the next millennium, Proceedings of the Ninth National Conference on System Dynamics. December 26 to 29, 2000, NAARM, Hyderabad.

Rao, K.H. 2000. High pressure technology in dairy processing. In: IX compendium on new concepts in dairy Technology, National Dairy Research Institute, Karnal : 207-212.

Rao, K.H. 2000. New concepts in concentration and drying of milk and milk products. in: IX compendium on new concept in dairy technology. National Dairy Research Institute, Karnal: 184-191.

Rao, N.H., Katyal, J.C., and Reddy, M.N. 2001. Embedding the sustainability perspective into agricultural research - implications for agricultural research management, *Journal of Food Policy* (accepted)

Publications

- Satyanarayana Rao, R.V. 2000. Evaluation of pigeonpea lines for antibiosis to *Helicoverpa armigera*. Indian Journal of Entomology 62(3): 312-314.
- Satyanarayana Rao, R.V. 2000. Resistance to leaf miner *Aproaerema modicella* (Deventer) in groundnut *Arachis hypogaea*. Indian Journal of Entomology 62(3): 280-285.
- Singh, R.R.B., Rao, K.H., Anjaneyulu, A.S.R., and Patil, G.R. 2000. Moisture Sorption properties of smoked chicken sausages from spent hen meat. Food Research International, 10: 1-6.
- Venkateswarlu, B., Mukhopadhyay, J., and Katyal, J.C. 2001. Selection of plus trees of neem for azadirachtin content and development of micropropagation protocol for mass propagation. In role of biotechnology in medicinal and aromatic plants (Vol. IV), Irfan A. Khan et al. (eds), Ukaaz Publications, India, pp 182-196.

12.2 Books / Reports

- Jagannadham Challa and Paul T. Perrault. 2000. Communication directory of national agricultural research system, NAARM, Hyderabad.
- Katyal, J. C. Sabarathnam, V.E., Sandhya Shenoy, N., Sontakki, B. S., Jayaraghavendra Rao V. K., and Soam, S.K. 2000. The reaction of Andhra Pradesh farmers towards the proposed agri-clinics and agri-business centres. NAARM, Hyderabad.
- Katyal, J. C., Sabarathnam, V.E., Sandhya Shenoy, N., Sontakki, B. S., Jayaraghavendra Rao V. K., and Soam, S. K. 2000. The reaction of farm graduates of Andhra Pradesh towards the proposed agri-clinics and agri-business centres. NAARM, Hyderabad.
- Kumar, P. 2000. Native traditional wisdom for livestock management, NAARM, Hyderabad.
- Rama Rao, D (Ed.) 2000. Application of system dynamics for development in the next millennium. Proceedings of the ninth national conference on system dynamics, December 26-29, 2000, NAARM, Hyderabad.
- Rama Rao, D., Kalpana Sastry, R., and Kiresur, V. 2000. Technological forecasting of future oilseeds scenario in India (ICAR-AP Cass Project) NAARM, Rajendranagar, Hyderabad, 271 pages.
- Sharma, G.D., Irfan Riazvi, and Rama Rao, D. 2001. Consultancy report on assessment of higher education in Andhra Pradesh, Submitted to DFID, New Delhi, March 2001.

12.3 Resource papers (developed and used as input in training)

- Human resource development through training (P. Manikandan)
- Book review (P. Manikandan)
- Performance appraisal of teachers (P. Manikandan)
- Brainstorming session for research project management (Jagannadham Challa)
- Economic valuation of biodiversity (B.S.Chandel)
- Impact assessment of agricultural R & D (B.S. Chandel)
- Agricultural Research - Emerging perspective in the context of new economic reforms (B.C.Chandel)
- FDP on educational psychology and behavioral technology (A. Gopalam, J.C.Challa and R.Kalpana Sastry)
- E-book presentation on "Tribal costume of Northern Karnataka" for Workshop on "Multimedia production and distribution" AIT and Eto, Bangkok, Thailand (Shailaja Naik and R.Kalpana Sastry)
- Powerpoint tutorial for use on online interactive teaching with tests of evaluation intermittently (R.Kalpana Sastry and Shailaja Naik)
- An approach to integrated farming system by Thai Farmer – A 20 minute presentation developed by video production processes as video cassette and VCD. (R. Kalpana Sastry, M.N. Reddy, R.Verma. and Raj Kamal)
- Hands-on Acrobat. A tutorial developed for preparation of 'e-book' (R. Kalpana Sastry)
- Towards sustainable agriculture - Slide tape module of 22 slides (R. Kalpana Sastry, M.N. Reddy, R.Verma and Raj Kamal)
- Water security in the twenty first century: Framework for research policy and action (N.H. Rao, J.C. Katyal and M.N. Reddy)
- Generating high research performance in ICAR: Reorienting personnel policy towards team performance and developing people - Paper prepared for M.V.Rao Committee (D. Rama Rao and N.H. Rao)
- Prioritizing the field level project options (S.K. Soam)
- Sustainability indicators for grassland management - A case study (S.K. Soam)
- Agroecosystem analysis (Transect, agroecosystem maps, bioresource flow diagrams, seasonal calendars and seasonal analysis) (S.K. Soam)
- Stakeholder analysis (S.K. Soam)

Publications

- ❑ Participatory rural appraisal techniques - Social Map, Mobility Map, Venn Diagramme, Time Line & Time Trend (B. S. Sontakki)
- ❑ Strategic research extension plan for research prioritization (V. K. Jayaraghavendra Rao and B. S. Sontakki)
- ❑ Present status of agricultural research extension linkages in India (B. S. Sontakki)
- ❑ Behavioural dimensions of linkages (V. K. Jayaraghavendra Rao & B. S. Sontakki)

13. Other Activities

13.1 Official Language Implementation

The official language policy of Government of India and instructions received from the ICAR were meticulously followed by the Hindi Cell and the following activities were carried out during the period under report. The correspondence in English and Hindi format are given in the following table. The proscribed levels were surpassed by the initiatives and decisions taken by the official language committee.

S.No	Period	Correspondence in English and Hindi	Correspondence in English alone
1	Jan – March 2000	1079	1802
2	April - June 2000	2271	1167
3	July - August 2000	1921	953
4	September - December 2000	1831	1230
	Total	7102	5152

- ✧ The tri-monthly meetings were held in March, June, October and December 2000 with nominated members, and the agenda notes was prepared before the meeting and the proceedings were circulated to the members incorporating the suggestions.
- ✧ The participants of foundation course on agricultural research management had an option to have the lecture notes in Hindi.
- ✧ The participants of the foundation course were given exposure on the official language policy guidelines and were apprised of the provisions for effective implementation of the official language policy. Apart from this presentation a capsule based training on functional Hindi was provided to the participants offering functional Hindi as an elective.

13.2 Celebration of Hindi Varsh

Competitions in Hindi were conducted in essay writing, elocution, one minute, translation of words, memory, making the words, dictation, noting and drafting, singing, and general knowledge. Participants excelled in the performance were awarded prizes. Hindi day was celebrated on September 14, 2000. Dr Shashi Mudiraj Professor and Head of the Hindi Department, University of Hyderabad delivered key note address and Dr J.C.Katyal Director NAARM, Hyderabad presided over the function.

14. Personnel

Director's Office

J.C. Katyal, Director
T.V. Ramdas, Personal Assistant
T. Vanisri, Personal Assistant

Joint Director's Office

S.N. Saha, Joint Director
Sarada Samanta, Personal Assistant

Agricultural Research Systems and Policies Management Division

T. Balaguru, Principal Scientist
K.P.C. Rao, Principal Scientist
S. Shanmugam, Principal Scientist
K.M. Reddy, Principal Scientist
M.M. Anwar, Principal Scientist
D. Rama Rao, Principal Scientist
N. Hanumantha Rao, Principal Scientist
Janardan Jee, Principal Scientist
R. Kalpana Sastry, Senior Scientist
B.S. Chandel, Senior Scientist
S.K. Nanda, Senior Scientist
S.K. Soam, Senior Scientist

Human Resource Development Division

P. Manikandan, Principal Scientist
A. Gopalam, Principal Scientist
Jagannadham Challa, Principal Scientist
K. Hanumantha Rao, Senior Scientist
R.V.S. Rao, Senior Scientist

Information and Communication Management Division

VE. Sabarathnam, Principal Scientist
K. Vidyasagar Rao, Principal Scientist
M. Narayana Reddy, Principal Scientist
N. Sandhya Shenoy, Senior Scientist
B.S. Sontakki, Senior Scientist
V.K.J.R. Rao, Senior Scientist

Training Cell

S. N. Saha, In-charge
P. Vijender Reddy, Technical Officer

Administration & Finance

Capt. Mehar Singh, CAO
B. Pandu Reddy, F&AO
K. Janakiramiah, AAO
Y. Shankar Rao, AAO

Computer Section

M.N. Reddy, In-Charge (ARIS lab)
N.H. Rao, In-charge (Computer lab)
M.J. Pradeep Kumar, Technical Officer
K.V. Kumar, Technical Officer

Farm Services

R. Kalpana Sastry, In-Charge
M.A. Basith, Technical Officer

Health Centre

A. Debnath, Medical Officer

Hostel Services

S. Shanmugam, In-Charge
Zameer Ahmed, Manager (Hostel Services)

Library Services

P. Manikandan, In-Charge
P.V. Nirmala, Technical Officer

Maintenance Section

V. Murali, In-Charge
Sohail Ahmed Khan, Technical Officer

Official Language Cell

A. Gopalam, In-Charge
D. Venkateswarlu, Technical Officer
J. Renuka, Asst. Director (OL)
S. Pradeep Singh, Asst. Director (OL)

Photography Section

N. Sandhya Shenoy, In-Charge
L. Venkateswarlu, Technical Officer

Press and Publication Services

D. Rama Rao, In-Charge
R.V.V.S. Prakasa Rao, Editor
Abhijit Bhattacharjee, PRO (On study leave)
N. Narasimha Reddy, Technical Officer
P. Namdev, Technical Officer

Security Cell

B.Ch. Satyanarayana, Estate-cum-Security Officer

Transport Services

S.K. Nanda, In-Charge
N.R. Nageswara Rao, Technical Officer

Personnel

Video Services

Janardan Jee, In-Charge
K.R. Prabhakar, Programme Officer (CCTV)
Ch. Janardhan Rao, Technical Officer

New Appointments

Dr R. V. Satyanarayana Rao, Senior Scientist on April 3, 2000.
Dr S.K. Soam, Senior Scientist on May 1, 2000
Dr B.S. Sontakki, Senior Scientist on May 22, 2000
Dr V.K.J. Rao, Senior Scientist on May 29, 2000

Promotions

S.No.	Name	From	To
1.	P. Vijender Reddy	T-4	T-5
2.	Sohail A Khan	T-4	T-5
3.	K.V. Kumar	T-4	T-5
4.	B. Veeraiiah	T-3	T-4
5.	N. Nareesh Kumar	T-3	T-4
6.	M. Shokhar Reddy	T-3	T-4
7.	S.K. Peera	T-3	T-4
8.	D. Rajagopal Rao	T-2	T-3

Farewell

- A fond farewell was given to Dr R. Chittiraichelvan, Principal Scientist of the Academy, by the faculty members of NAARM on his new assignment as Head, CHES, Gonikoppal of IIHR, Bangalore. He was relieved on November 5, 2000 after a stint of 13 years of service at the Academy.
- NAARM family bid farewell to Dr Ajit P Maru, Senior Scientist of the Academy on August 28, 2000 to join as Research Officer, ISNAR, The Netherlands.

The Academy wishes them success in their respective new assignments.

Retirement

- Dr P. Kumar, Principal Scientist and Head, Agricultural Research Systems and Policies Management Unit, retired on superannuation on September 30, 2000, after twelve years of service at the Academy. NAARM family wishes a happy retired life to Dr Kumar.
- Mr A.M. Venkateswarlu, Pharmacist, retired on superannuation on December 31, 2000. NAARM family wishes him a happy retired life.

अप्रैल 2000

छाटा विश्लेषण हेतु सांख्यिकीय साफ्टवेयर पर एक दस दिवसीय कार्यक्रम का आयोजन 4 से 15 अप्रैल, 2000 तक किया गया। इसमें भा.कृ.अनु.प. के विभिन्न संस्थानों के ग्यारह वैज्ञानिक और तकनीकी अधिकारियों ने भाग लिया। डॉ. के. विद्यासागर राव और डॉ. य. एम. नारायण रेड्डी इस कार्यक्रम के समन्वयक थे।

तीसरा वार्षिक अनुसंधान संबंध विकास कार्यक्रम 24-29 अप्रैल 2000 तक आयोजित किया गया। इसमें भा.कृ.अनु.प. संस्थानों के कुल 18 विभागाध्यक्ष और परियोजना समन्वयकों ने भाग लिया। डॉ. जगन्नाधम चेल्ला और डॉ. एस.एन. साहा ने कार्यक्रम का समन्वयन किया।

तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के वरिष्ठ संकाय सदस्यों हेतु शिक्षा मनोविज्ञान और आचरण प्रौद्योगिकी पर संकाय विकास कार्यक्रम अप्रैल 10 से 20, 2000 तक आयोजित किया गया। 18 संकाय सदस्यों ने इस कार्यक्रम में भाग लिया। डॉ. ए. गोपालम, डॉ. जगन्नाधम चेल्ला और डॉ. आर. कल्पना शास्त्री इसके समन्वयक थे।

तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक और वित्तीय कर्मियों हेतु शैक्षिक प्रशिक्षण और वित्तीय मुद्दों का प्रबंध पर एक दिनाभिन्यास कार्यक्रम 29 अप्रैल से 09 मई, 2000 तक आयोजित किया गया। इस कार्यक्रम द्वारा 27 सहभागी लाभान्वित हुए। इसके समन्वयक डॉ. पी.सी. राव, डॉ. ए. गोपालम और डॉ. डी. रामाराव थे।

तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के संकाय सदस्यों हेतु कृषि अनुसंधान और शिक्षा में कंप्यूटर उपयोग पर एक प्रायोजित कार्यक्रम अप्रैल 29 से मई 04, 2000 तक आयोजित किया गया। इसमें कुल 17 सदस्यों ने भाग लिया। डॉ. एन. हनुमंत राव, डॉ. आर. चित्तरंजनेल्वन और डॉ. अजित मारु इस कार्यक्रम के समन्वयक थे।

तंजानिया के लिए कृषि अनुसंधान प्रबंध का अन्तर्राष्ट्रीय कार्यक्रम 24 अप्रैल से 02 जून, 2000 तक आयोजित किया गया। इसमें तंजानिया के नौ सदस्यों ने भाग लिया इसके समन्वयक डॉ. टी. बालगुरु थे।

मई 2000

महिला वैज्ञानिकों के लिए एक सप्ताह का प्रबंध कार्यक्रम 1 से 6 मई, 2000 तक आयोजित किया गया। इसमें 17 महिला वैज्ञानिकों ने भाग लिया। डॉ. जनार्दनजी तथा डॉ. एन. संच्या शेणार्थ ने कार्यक्रम का समन्वयन किया।

संरक्षित कृषि पर दो दिवसीय ए.एच.आर.डी. कार्यक्रम (सेमिनार) 10-11 मई, 2000 तक आयोजित किया गया। इसमें रा.कृ.वि. और भा.कृ.अनु.प. एवं राज्य सरकार विभाग एवं निजी क्षेत्र के कुल 60 वैज्ञानिकों ने भाग लिया। डॉ. डी. रामाराव इस कार्यक्रम के समन्वयक थे। प्रो. मेड्डी एच. जेनसेन, अरिजोना विश्वविद्यालय और विश्व बैंक, परामर्शदाता इसके संसाधन व्यक्ति थे।

शिक्षा मनोविज्ञान आचरण प्रौद्योगिकी पर एक सप्ताह का संकाय विकास कार्यक्रम तमिलनाडु कृषि विश्वविद्यालय के अध्यापक और वरिष्ठ स्तर के प्रबंधकों हेतु 29 मई से 03 जून, 2000 तक आयोजित किया गया। डॉ. ए. गोपालम, डॉ. पी. मणिकण्डन और डॉ. पी. कुमार ने इस कार्यक्रम का समन्वयन किया।

भा.कृ.अनु.प. में प्रशासनिक प्रबंध सुधार पर दस दिवसीय कार्यक्रम 30 मई से 09 जून, 2000 तक आयोजित किया गया। इसमें भा.कृ.अनु.प. संस्थानों के 26 प्रशासनिक व वित्त अधिकारियों ने भाग लिया। इसका समन्वयन डॉ. एस.एन. साहा और श्रीमती. पाण्डु रेड्डी ने किया।

जून 2000

- * कृषि में कंप्यूटर प्रयोग पर दस दिवसीय कार्यक्रम 2 जून से 01 जुलाई, 2000 तक आयोजित किया गया। भा.कृ.अनु.प. और रा.कृ.वि. से 14 वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। डॉ. के. विद्यासागर राव और डॉ. के. एम. रेड्डी सि कार्यक्रम के समन्वयक थे।
- * कृषि विस्तार और अनुसंधान में शीघ्र, विश्रान्त और सहभागी ग्रामीण मूल्यांकन और सहवागी अधिगम और कार्यतकनीक पर 20 जून से 01 जुलाई, 2000 तक दस दिवसीय कार्यक्रम का आयोजन किया गया। इसमें कुल 17 लोग जिनमें भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के वैज्ञानिक वृत्ति दिवास फेंद्र और वृत्त्युप प्रशिक्षण केंद्र के प्रशिक्षक सम्मिलित थे, ने भाग लिया। डॉ. वीई सबारत्नम और डॉ. एन. संध्या शेणाय्य इसके समन्वयक थे।
- * तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर के संकाय सदस्यों हेतु शिक्षा मनोविज्ञान और आचरण प्रौद्योगिकी पर दस दिवसीय संकाय विकास कार्यक्रम 26 जून से 06 जुलाई, 2000 तक आयोजित किया गया। इसमें 22 सहभागियों ने भाग लिया। डॉ. ए. गोपालम, डॉ. मणिकण्डन और डॉ. एम. शणमुगम ने इस आयोजित कार्यक्रम का समन्वयन किया।

जुलाई 2000

- * भा.कृ.अनु.प. संस्थानों के लिए राजभाषा नीति पर पाँच दिवसीय प्रशिक्षण व कार्यशाला का आयोजन 3-7 जुलाई, 2000 कर किया गया। इस कार्यशाला में 38 अधिकारी/कर्मचारियों ने भाग लिया। डॉ. ए. गोपालम और श्री प्रदीप सिंह ने इसका समन्वयन किया।
- * कृषि अनुसंधान प्रबंध का तीसरा कार्यपालक विकास कार्यक्रम 15-18 जुलाई, 2000 तक आयोजित किया गया। इसमें भा.कृ.अनु.प. संस्थानों से 12 विभागाध्यक्ष एवं निदेशकों ने भाग लिया। डॉ. एस.एन. साहा और डॉ. जगन्नाथम चेल्ला इस कार्यक्रम के समन्वयक थे।
- * व्यक्तित्व और संगठनात्मक प्रभावत्मकता हेतु कृषि वैज्ञानिक विकास पर एक कार्यक्रम 17-28 जुलाई, 2000 तक आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के ग्यारह वैज्ञानिक ने इसमें भाग लिया। डॉ. एस. शणमुगम और डॉ. जगन्नाथम चेल्ला ने कार्यक्रम का समन्वयन किया।
- * शैक्षिक वीडियो निर्माण पर दो सप्ताह का कार्यक्रम 18-29 जुलाई, 2000 तक आयोजित किया गया। इसमें भा.कृ.अनु.प. संस्थानों से 4 वैज्ञानिक और तकनीकी अधिकारियों ने भाग लिया। डॉ. आर.सी. चेल्वन और श्री के. आर. प्रभाकर ने इसका समन्वयन किया।
- * प्रभावी प्रस्तुतीकरण हेतु शैक्षिक उपकरण पर दो सप्ताह का कार्यक्रम 18-29 जुलाई, 2000 तक आयोजित किया गया। इसमें दो सदस्यों ने भाग लिया। डॉ. एन. संध्या शेणाय्य एवं डॉ. जनार्दन जी ने इस कार्यक्रम का समन्वयन किया।
- * कृषि अनुसंधान और शिक्षा में कंप्यूटर उपयोग पर एक सप्ताह का कार्यक्रम 17-22 जुलाई, 2000 तक आयोजित किया गया। इसमें पंद्रह संकाय सदस्यों ने भाग लिया। डॉ. एन. हनुमंत राव, डॉ. के. विद्यासागर राव और डॉ. आर.सी. चेल्वन इस कार्यक्रम के समन्वयक थे।
- * तमिलनाडु कृषि विश्वविद्यालय के प्रशासन एवं वित्त खंड के संकाय सदस्यों हेतु शिक्षा प्रशासन और वित्तीय मुद्दों का प्रबंध पर एक दिग्बिन्ध्यास कार्यक्रम 17-29 जुलाई, 2000 तक आयोजित किया गया। इसमें 27 सदस्यों ने भाग लिया। डॉ. के.पी.सी. राव डॉ. जनार्दन जी, और डॉ. ए. गोपालम इस कार्यक्रम के समन्वयक थे।

अगस्त 2000

- कृषि संपोष मूल्यांकन के लिए विश्लेषणात्मक उपकरण एवं तकनीक पर 01-10 अगस्त, 2000 तक एक दस दिवसीय कार्यक्रम आयोजित किया गया। इस कार्यक्रम में सात वैज्ञानिकों ने भाग लिया। डॉ. एन. हनुमंत राव एवं डॉ. एम. नारायण रेड्डी ने इस कार्यक्रम का समन्वयन किया।
- 21-26 अगस्त, 2000 के दौरान कृषि अनुसंधान प्राथमिकता तकनीक पर एक सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में इकतीस सदस्यों ने भाग लिया। डॉ. के.पी.सी. राव एवं डॉ. बी.एस. चन्देल ने इस कार्यक्रम का समन्वयन किया।

सितम्बर 2000

- कृषि अनुसंधान एवं विकास में सहभागिता पद्धति पर 11-16 सितम्बर, 2000 के दौरान एक सप्ताह का कार्यक्रम आयोजित किया। भा.कृ.अनु.प. संस्थानों के सात वैज्ञानिकों ने इस कार्यक्रम में भाग लिया। डॉ. एन. संध्या शेणार्थ एवं डॉ. बीई सवारत्न ने इस कार्यक्रम का समन्वयन किया।
- नये पर्यावरण में सफलता तथा नेतृत्व के अध्ययन पर 14 से 20 सितम्बर, 2000 के दौरान एक सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के चार वैज्ञानिकों ने भाग लिया। डॉ. जनार्दनजी एवं डॉ. जगन्नाथम थल्ला ने इस कार्यक्रम का समन्वयन किया।
- कंप्यूटर की जानकारी पर 18-23 सितम्बर, 2000 के दौरान एक कार्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के 19 वैज्ञानिक एवं तकनीक अधिकारियों ने इस कार्यक्रम में भाग लिया। डॉ. के.एम. रेड्डी इस कार्यक्रम के समन्वयक रहे।
- वीडियो संपादन तकनीक पर दस दिवसीय कार्यक्रम 19-29 सितम्बर, 2000 तक आयोजित किया गया। इस कार्यक्रम में विभिन्न अनुसंधान संस्थानों के इस कार्यक्रम में कार्यरत छह तकनीकी कर्मियों ने भाग लिया। श्री के.आर. प्रभाकर का समन्वयन किया।
- कृषि अनुसंधान एवं विकास के प्रभावात्मक निर्धारण पर एक सप्ताह का कार्यक्रम 25-30 सितम्बर, 2000 के दौरान आयोजित किया गया। भा.कृ.अनु.प. संस्थानों तथा कृषि राज्य विश्वविद्यालयों के 15 वैज्ञानिक, जिन्हें, विभिन्न गणितीय-आर्थिक मापन तकनीकों का ज्ञान है, ने इस कार्यक्रम में भाग लिया। डॉ. बी.एस. चन्देल एवं डॉ. के.पी.सी. राव ने इस कार्यक्रम का समन्वयन किया।
- निर्णय समर्थन पद्धति पर 25 सितम्बर 15 अक्टूबर, 2000 के दौरान एक कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों एवं राज्य कृषि विश्वविद्यालयों के 21 वैज्ञानिकों एवं संकाय सदस्यों ने भाग लिया। डॉ. डी. रामाराव एवं डॉ. एस.के. नन्दा इस कार्यक्रम का समन्वयन किया।
- शिक्षा मनोविज्ञान एवं व्यवहारिक प्रौद्योगिकी पर एक दस दिवसीय संकाय विकास कार्यक्रम विशेषतः तमिलनाडु कृषि विश्वविद्यालय के शिक्षकों तथा वरिष्ठ स्तर के प्रबंधकों के लिए 11-21 सितम्बर, 2000 के दौरान आयोजित किया गया। डॉ. टी. बालगुरु ने इस कार्यक्रम का समन्वयन किया।
- भारतीय कृषि सांख्यिकीय अनुसंधान संस्थान द्वारा 27-29 सितम्बर, 2000 के दौरान परियोजना सूचना प्रबंध पर तीन दिवसीय प्रशिक्षण व कार्यशाला आयोजित की गई। इस कार्यक्रम में 10 वैज्ञानिकों ने भाग लिया। डॉ. एम. नारायण रेड्डी ने इस कार्यक्रम का समन्वयन किया।

अक्टूबर 2000

- * माइक्रोसॉफ्ट परियोजना के प्रयोग द्वारा परियोजना प्रबंध पर 30 अक्टूबर से 4 नवम्बर, 2000 तक एक सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 15 सदस्यों ने भाग लिया। डॉ. डी. रामाराव इस कार्यक्रम के समन्वयक रहे।

नवम्बर 2000

- * कृषि अनुसंधान प्रबंध पर एक अन्तर्राष्ट्रीय कार्यक्रम कृषि निदेशकों, येमन रिपब्लिक हेतु 01 नवम्बर से 30 दिसम्बर, 2000 के दौरान आयोजित किया गया। डॉ. टी. बालगुरु इस कार्यक्रम के समन्वयक रहे।
- * कृषि में भौगोलिक सूचना पद्धति प्रयोग पर 6-16 नवम्बर, 2000 के दौरान एक दस दिवसीय कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 13 सदस्यों ने भाग लिया। डॉ. एन. हनुमंत राव एवं डॉ. एम.एन. रेड्डी ने इस कार्यक्रम का समन्वयन किया।
- * तमिलनाडु कृषि विश्वविद्यालय कोयंबतूर के वरिष्ठ स्तर के प्रबंधकों के लिए शैक्षणिक प्रशासन एवं वित्तीय मामलों के प्रबंध पर नवम्बर 7 से 17, 2000 तक पुनर्गठनमुखीकरण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 26 सदस्यों ने भाग लिया। डॉ. गोपालम, कैप्टन मेहर सिंह एवं श्री बी. पाण्डु रेड्डी इस कार्यक्रम का समन्वयक रहे।
- * कृषि अनुसंधान परियोजना प्रबंध पर एक दस दिवसीय कार्यक्रम 14 से 24 नवम्बर, 2000 के दौरान आयोजित किया गया। इस कार्यक्रम में ग्यारह सदस्यों ने भाग लिया। डॉ. एस.के. नन्दा एवं डॉ. डी. रामाराव ने इस कार्यक्रम में समन्वयन किया।
- * तमिलनाडु कृषि विश्वविद्यालय सदस्यों के लिए कृषि अनुसंधान व शिक्षा के लिए कंप्यूटर प्रयोग पर दो प्रशिक्षण कार्यक्रम क्रमशः 13 से 18 नवम्बर, 2000 तथा 18 से 23 नवम्बर, 2000 के दौरान आयोजित किये गये। प्रथम कार्यक्रम का डॉ. के. विद्यासागर राव, डॉ. एन. हनुमंत राव व डॉ. एम. नारायण रेड्डी तथा दूसरा कार्यक्रम का डॉ. डी. रामाराव, डॉ. एन. हनुमंत राव एवं डॉ. आर. कल्पना शास्त्री द्वारा समन्वयन किया गया।
- * भारतीय बीज उद्योग प्रबंध पर नवम्बर 20 से 25, 2000 के दौरान एक सप्ताह का कार्यक्रम आयोजित किया गया। 22 सदस्यों ने इस कार्यक्रम में भाग लिया जिसका समन्वयन डॉ. के.पी.सी. राव एवं डॉ. जनार्दन जी ने किया।
- * नयी सहस्राब्दि राजभाषा नीति संकल्प पर 28-29 नवम्बर, 2000 के दौरान एक कार्यशाला आयोजित की गई 47 सदस्यों ने इस कार्यशाला में भाग लिया। डॉ. गोपालम एवं श्री डी. वेंकटेश्वर्तु ने इस कार्यशाला का समन्वयन किया।
- * अनुसंधान व प्रबंध के उन्नत डाटा विश्लेषण तकनीक पर एक दस दिवसीय पाठ्यक्रम 28 नवम्बर से 8 दिसम्बर, 2000 के दौरान आयोजित किया गया। इस पाठ्यक्रम में आठ सदस्यों ने भाग लिया। डॉ. एम. नारायण रेड्डी और डॉ. के. विद्यासागर राव ने इस कार्यक्रम का समन्वयन किया।
- * कृषि अनुसंधान में चौथा प्रबंध विकास कार्यक्रम 30 नवम्बर से 6 दिसम्बर, 2000 के दौरान आयोजित किया गया। भा.क.अनु.प. संस्थान के 13 प्रभागध्यक्षों ने इस कार्यक्रम में भाग लिया। डॉ. जगन्नाथम चेल्ला एवं डॉ. एस.एन. साहा ने इस कार्यक्रम का समन्वयन किया।

दिसम्बर 2000

- * कृषि अनुसंधान संगठन में उत्तम शासन पर एक दिवसीय मस्तिष्क मंथन सत्र 02 दिसम्बर, 2000 को आयोजित किया गया। राष्ट्रीय कृषि विज्ञान अकादमी तथा नार्म संकाय क 19 सदस्यों ने इस सत्र में भाग लिया डॉ. जे.सी. कल्याल एवं डॉ. एन. हनुमंत राव ने इस सत्र का समन्वयन किया।
- * भा.कृ.अनु.प. संस्थानों के प्रसासन एवं वित्तीय प्रबंध के सुधार पर दस एक दिवसीय कार्यक्रम 6 से 16 दिसम्बर, 2000 के दौरान आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. के विभिन्न संस्थानों से 17 प्रशासनिक एवं वित्तीय अधिकारियों ने भाग लिया। डॉ. एस.एन.साह, एवं कैप्टन सहा सिंह, श्री जी. नयडु रेड्डी ने इस कार्यक्रम का समन्वयन किया।
- * 12 से 23 दिसम्बर, 2000 के दौरान कृषि में कंप्यूटर प्रयोग पर दो सप्ताह का कार्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के नौ वैज्ञानिकों ने इस कार्यक्रम में भाग लिया डॉ. के.एम. रेड्डी एवं डॉ. के. विद्यासागर राव ने इस कार्यक्रम का समन्वयन किया।
- * शिक्षण वीडियो उत्पादन पर 12-23 दिसम्बर, 2000 तक दो सप्ताह का कार्यक्रम आयोजित किया गया। विभिन्न भा.कृ.अनु.प. संस्थानों के तीन तकनीकी अधिकारियों ने इस कार्यक्रम में भाग लिया। श्री के.आर. प्रभाकर एवं डॉ. जनार्दन जी इस कार्यक्रम के समन्वयक रहे।
- * 28-29 दिसम्बर, 2000 के दौरान सिस्टम्स डॉ. यनाम्लिस सोसाइटी ऑफ इण्डिया के तत्वावधान में कृषि विकास के लिए सिस्टम्स डॉ. यनाम्लिस पर दो दिवसीय राष्ट्रीय सम्मेलन आयोजित किया गया। सिस्टम डॉ. याम्लिस पर 26-27 दिसम्बर, 2000 के दौरान सम्मेलन पूर्व प्रशिक्षण आयोजित किया गया। इस सम्मेलन में भा.कृ.अनु.प., आई.आई.टी., टाटा परामर्श सेवा, मैनेज एवं अन्य संगठन जिसमें कृषि विश्वविद्यालय, परामर्श संगठन इत्यादि से वैज्ञानिक एवं अधिकारी प्रतिनिधि सम्मिलित थे। डॉ. डॉ. रामाराव इस सम्मेलन के समन्वयक रहे।

जनवरी 2001

- * कृषि प्रौद्योगिकी सूचना केंद्रों के वेब पेजस की अभिकल्पना तथा विकास पर 17 से 19 जनवरी, 2001 के दौरान तीन दिवसीय प्रशिक्षण कार्यक्रम आयोजित किया गया। कृषि प्रौद्योगिकी सूचना केंद्रों के 20 वैज्ञानिकों ने इस कार्यक्रम में भाग लिया। डॉ. एन. संध्या शेणार्थ एवं डॉ. एम.एन. रेड्डी इस कार्यक्रम के समन्वयक रहे।
- * कृषि अनुसंधान प्राथमिकता तकनीकी पर एक सप्ताह का पाठ्यक्रम 18-24 जनवरी, 2001 के दौरान आयोजित किया गया। 15 सदस्यों ने इस पाठ्यक्रम में भाग लिया। डॉ. के.पी.सी.राव एवं डॉ. बी.एस.चन्देल ने इस पाठ्यक्रम का समन्वय किया।
- * गहन हिन्दी प्रशिक्षण व कार्यशाला 22-27 जनवरी, 2001 तक आयोजित की गयी। डॉ. ए. गोपालम एवं श्री मती जे. रेणुका इसकार्यशाला के समन्वयक रहे।
- * श्रीलंका के प्रशासकों एवं लेखाकार के लिए एक अन्तर्राष्ट्रीय कार्यक्रम 22-27 जनवरी, 2001 तक आयोजित किया गया। इस कार्यक्रम में श्रीलंका के सात प्रशासकों एवं लेखाकारों ने भाग लिया। डॉ. टी. बालगुरु, डॉ. आर. कल्पना शास्त्री एवं डॉ. एम.एन. रेड्डी ने इस कार्यक्रम का समन्वयन किया।

फरवरी 2001

- * कृषि अनुसंधान प्रबंध पर एक कार्यपालक विकास कार्यक्रम 2 से 5 फरवरी, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के छह विभागाध्यक्षों एवं निदेशकों ने भाग लिया। डॉ. एस.एन. साहा एवं जगन्नाथम चेल्ला ने इस कार्यक्रम का समन्वयन किया।
- * तमिलनाडु कृषि विश्वविद्यालय के संकायाध्यक्षों एवं निदेशकों के लिए नेतृत्व एवं प्रबंध पर 8-10 फरवरी, 2001 के दौरान तीन दिवसीय कार्यशाला आयोजित की गई। इस कार्यशाला में 22 सदस्यों ने भाग लिया जिसके एवं डॉ.पी. गणिवण्ड्य रहे।
- * कृषि में कंप्यूटर प्रयोग पर 13-23 फरवरी, 2001 के दौरान दस दिवसीय पाठ्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के 10 वैज्ञानिकों एवं तकनीकी अधिकारियों ने इस कार्यक्रम में भाग लिया। डॉ. के. विद्यासागर राव एवं डॉ. के.एम. रेड्डी कार्यक्रम के समन्वयक रहे।
- * श्रीलंका के वैज्ञानिकों के लिए कृषि अनुसंधान प्रबंध पर दो सप्ताह का अन्तर्राष्ट्रीय कार्यक्रम 22 फरवरी से 7 मार्च, 2001 के दौरान आयोजित किया गया। श्रीलंका के तीन वैज्ञानिकों ने इस कार्यक्रम में भाग लिया। डॉ. एस.एन. साहा, डॉ.एस. शण्मुगम एवं डॉ.एस.के. नन्दा इस कार्यक्रम के समन्वयक रहे।

मार्च 2001

- * श्रीलंका के वैज्ञानिकों हेतु कृषि अनुसंधान प्रबंध पर तीन सप्ताह का अन्तर्राष्ट्रीय कार्यक्रम 12-31 मार्च, 2001 तक आयोजित किया गया। श्रीलंका के 10 वैज्ञानिकों ने इसमें भाग लिया। डॉ. डी. रामाराव, डॉ. जनार्दन जी और डॉ. बी.एस. चन्देल इस कार्यक्रम के समन्वयक थे।
- * कृषि में प्रभावी संपर्क तकनीक पर एक सप्ताह का पाठ्यक्रम 14-21 मार्च, 2001 तक आयोजित किया गया। विभिन्न भा.कृ.अनु.प. संस्थानों के सात वैज्ञानिकों ने इसमें भाग लिया। डॉ. वीई. सबारत्नम, डॉ. वी.के.जे.आर. राव और डॉ. बी.एस. सौनटक्की इस पाठ्यक्रम के समन्वयक थे।
- * महिला वैज्ञानिकों हेतु एक सप्ताह का प्रबंध कार्यक्रम 19-24 मार्च, 2001 तक आयोजित किया गया। इसमें 23 महिला वैज्ञानिकों ने भाग लिया। डॉ. जनार्दन जी और डॉ. एन. संथ्या शैणायें इस कार्यक्रम के समन्वयक थे।
- * डाटा विश्लेषण हेतु सांख्यिकीय साफ्टवेयर पर दस दिवसीय पाठ्यक्रम 20-31 मार्च, 2001 तक आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के 7 वैज्ञानिक और तकनीकी अधिकारियों ने इसमें भाग लिया। डॉ. के. विद्यासागर राव, डॉ. एम. नारायण रेड्डी इस पाठ्यक्रम के समन्वयक थे।
- * एस सप्ताह का कंप्यूटर आधारित शिक्षण प्रौद्योगिकी कार्यक्रम, तमिलनाडु कृषि विश्वविद्यालय के संकाय सदस्यों हेतु 2-9 मार्च, 2001 तक आयोजित किया गया। इसमें त.कृ.वि. के विभिन्न महाविद्यालयों के 24 संकाय सदस्यों ने भाग लिया। डॉ. ए. गोपालम डॉ. एन. हनुमंत राव और डॉ. एम. नारायण रेड्डी ने कार्यक्रम का समन्वयन किया।



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